Hamiltonians of mean force

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We study the operator representability problem for the Hamiltonian of mean force (HMF) in Gaussian linearly coupled environments beyond the commuting limit. Building on the quenched-density construction, we derive an exact imaginary-time reformulation in which bath statistics enter as scalar kernel moments while operator growth is generated by the adjoint chain $\{\mathrm{ad}^n_{H_Q}(f)\}_{n\geq 0}$. This yields a structural criterion: a finite closed-form HMF exists only when the adjoint-generated operator algebra closes inside the target ansatz. We then instantiate the criterion in a Pauli-closed qubit model and obtain an explicit closed-form mean-force Hamiltonian with temperature- and bath-dependence carried by mode-resolved kernel coefficients. The manuscript is framed as the noncommuting Gaussian follow-up to the commuting benchmark analysis of arXiv:2602.13146 and sets up the non-Gaussian extension program.

INTRODUCTION

This manuscript is a direct follow-up to Ref. [1] (arXiv:2602.13146). That paper established the quenched-density representation and delivered an exact benchmark in the commuting Gaussian sector. The present paper starts from that benchmark and addresses the next step: how the Hamiltonian of mean force is organised once the coupling operator does *not* commute with the bare system Hamiltonian.

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian: $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operationally defined equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\bar{\rho}_Q(\beta) = \mathrm{Tr}_X\, e^{-\beta H_{\mathrm{tot}}}, \tag{1}$$

$$e^{-\beta H_{\mathrm{MF}}(\beta)} \propto \frac{\mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \mathrm{Tr}_X e^{-\beta H_X}. \tag{2}$$

This object is generally not $e^{-\beta H_Q}$ for the bare system Hamiltonian H_Q. The resulting representational question is precise: what operator, if any, plays the role of an equilibrium generator for the subsystem once the coupling is non-negligible? The Hamiltonian of mean force (HMF) answers this by construction and is the standard starting point in strong-coupling thermodynamics and the mean-force Gibbs-state program [2–7].

Why care about a mean-force Hamiltonian at all? First, it provides the exact reduced equilibrium object that underlies strong-coupling thermodynamic identities, including free-energy and work relations that remain valid beyond weak coupling[2, 5, 8–11]. Second, it furnishes a consistent equilibrium initialization for open-system dynamics when correlations with the bath are unavoidable, a point emphasized in the literature on correlated initial states and reduced dynamics [4, 12]. Related subensemble approaches explicitly treat the subsystem thermodynamics as inherited from a global canonical ensemble, making the effective reduced generator central to the formalism[13]. In short, the HMF is not an optional reinterpretation; it is the exact operator that encodes the reduced equilibrium state whenever system–bath coupling is finite.

Historically, open-system theory prioritized weak-coupling and Markovian regimes, where reduced equilibrium can often be approximated by a Gibbs state of a renormalized H_Q. At finite coupling the reduced state inherits explicit temperature dependence and interaction-induced operator content that is not captured by a simple renormalization. Coupling-dependent thermodynamic response features in quantum Brownian motion and related models highlight this complexity[14, 15]. The strong-coupling literature consequently treats the mean-force Gibbs state as a distinct equilibrium object, with operational ramifications for heat and energy definitions[9, 16–18]. The HMF can encode interaction-induced operator content absent from H_Q [14, 15, 19], making explicit representation of this content a primary goal.

The HMF literature itself is broad but structured. Canonical definitions and thermodynamic identities are developed in strong-coupling thermodynamics and fluctuation-relation work[2, 3, 5, 8]. A comprehensive review consolidates the "static" mean-force Gibbs perspective with the "dynamical" return-to-equilibrium perspective in open quantum systems [4]. Operational questions such as measurability and thermodynamic consistency at strong coupling have also been pursued[9, 17, 18].

Exact or controlled evaluations exist in special cases. For commuting (QND) interactions, the operator algebra closes trivially and the HMF can be written explicitly[20]. Quadratic/Gaussian models (e.g., damped harmonic oscillators) are solvable because Gaussianity is preserved, leading to closed operator forms[21–23]. Finite-dimensional closures such as spin-boson or single-qubit models provide additional controlled benchmarks[24]. Beyond these cases, the mean-force Gibbs state is often accessed through systematic limits: Cresser and Anders derive weak- and ultrastrong-coupling expressions and

show that, in the ultrastrong limit, the mean-force Gibbs state becomes diagonal in the interaction basis rather than the system Hamiltonian basis[25, 26]. Recent work generalizes the HMF framework to finite baths by introducing a pair of quantum Hamiltonians of mean force that incorporate bath feedback[27], and structural studies further analyze the operator content and potential renormalization of the HMF in extended settings[19, 28].

Outside solvable models, most approaches are perturbative or numerical. Weak-coupling expansions yield controlled but limited series [25, 29]. Reaction-coordinate mappings[30–33] and polaron transforms provide approximate local generators applicable at intermediate coupling. Imaginary-time path-integral methods, stochastic representations, and hierarchical-equations techniques can compute $\bar{\rho}_Q(\beta)$ directly but typically do not yield compact operator forms for $H_{\rm MF}$[34–39]. Stochastic Liouville and partition-free approaches give complementary numerical access without yielding closed-form HMFs [40, 41]. These methods are indispensable for quantitative predictions but leave open the representational question: when does a local or otherwise restricted operator form exist?

The influence-functional formalism is a natural language for this problem. For Gaussian baths with linear coupling it provides an exact route to integrating out bath degrees of freedom[21, 22, 42]. In equilibrium it becomes a Euclidean (imaginary-time) influence functional, usually bilocal in τ, and admits a Hubbard–Stratonovich rewriting as a quenched Gaussian-field average[43–45]. Related path-integral derivations of quantum Langevin dynamics make explicit the noise and dissipation structure inherited from the bath[46, 47]. This formulation also clarifies the terminology: "nonlocal" initially refers to imaginary-time nonlocality of the kernel, whereas the HMF locality question concerns the operator structure on the system Hilbert space.

The existence of $H_{\rm MF}$ is therefore not the issue—it is defined by a logarithm of a traced exponential. The real obstruction is representability: when does $H_{\rm MF}$ admit a closed-form expression within a restricted operator family (few-body, spatially local, or a given algebra)? Much of the literature either (i) solves special models, (ii) expands in controlled limits, or (iii) computes $\bar{\rho}_Q(\beta)$ numerically and analyzes its properties, but a general structural criterion is still lacking [25, 27, 28].

Relative to Ref. [1], the present manuscript has a narrower but deeper scope. We retain the exact Gaussian influence-functional and quenched-field identities, then use adjoint-action and Magnus/BCH organisation [48–50] to study representability in the noncommuting sector. The central result is an explicit closure criterion: a finite closed-form HMF exists only when the operator algebra generated by repeated adjoint action of H_Q on the coupling operators closes inside the target ansatz. We then instantiate this criterion in a Pauli-closed qubit example, where all bath dependence is reduced to scalar kernel moments and the mean-force Hamiltonian is written in exact closed form. Moreover, it is not often one has the opportunity to say something new about a qubit.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [1]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\rm tot} = H_Q + H_X + H_{\rm int}, \tag{3}$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\rm int} = f \otimes B, \tag{4}$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv {\rm Tr}_X e^{-\beta H_{\rm tot}}$. As shown in Ref. [1], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi\big[U_\xi(\beta)\big], \tag{5}$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\big(H_Q + \xi(\tau) f\big)\right], \tag{6}$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f, its influence can be captured by attaching a τ-dependent driving field to f [51, 52]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But ${\rm Tr}_X$ averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau) f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is

required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \qquad \tau \in [0, \beta], \tag{7}$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X}/Z_X$. For an arbitrary c-number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \mathrm{Tr}_X \left[\mathcal{T}_\tau \exp\left(-\int_0^\beta d\tau\, j(\tau)\, B(\tau) \right) \rho_X \right]. \tag{8}$$

Here $j(\tau)$ is introduced as an external c-number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c-number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [21, 22, 42]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \qquad \Phi[f] \equiv \log \mathcal{F}[f]. \tag{9}$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected expansion [53, 54]

$$\begin{aligned} \Phi[f] = \sum_{n\geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ \times K^{(n)}(\tau_1, \ldots, \tau_n)\, f(\tau_1) \cdots f(\tau_n), \end{aligned} \tag{10}$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [55]

$$K^{(n)}(\tau_1, \ldots, \tau_n) \equiv \left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c. \tag{11}$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c = (-1)^n \left. \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \right|_{j=0}. \tag{12}$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n\geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau\, f(\tau) B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [43–45]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp\left(-\int_0^\beta d\tau\, \xi(\tau)\, f(\tau) \right) \right], \tag{13}$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ-histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c-number field, its n-point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle_c = K^{(n)}(\tau_1, \ldots, \tau_n). \tag{14}$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}\big(\{\tau_i\}_{i\in C}\big), \tag{15}$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \ldots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\left\langle \xi(\tau_1)\xi(\tau_2) \right\rangle = K^{(2)}(\tau_1, \tau_2), \tag{16}$$

$$\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3) \right\rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \tag{17}$$

$$\begin{aligned} &\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3)\xi(\tau_4) \right\rangle \\ &\quad = K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \tag{18}$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}, \qquad H_{\mathrm{MF}}(\beta) \;\equiv\; -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \tag{19}$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi[U_\xi(\beta)]\,. \tag{20}$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing

the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \tag{21}$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{m_k\omega_k}\delta(\omega - \omega_k)$ via the relation [56]

$$K(\tau) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}. \tag{22}$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned}\int_0^\beta d\tau\, K(\tau) &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\int_0^\beta d\tau\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}\\ &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{2}{\omega}\\ &= 2\lambda,\end{aligned} \tag{23}$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau\,\xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi[\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau - \tau') = 2\beta\lambda. \tag{24}$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ-independent in imaginary time. In this instance time ordering drops out, and the average is given by [1]:

$$\bar{\rho}_Q(\beta) = \exp\left[-\beta\left(H_Q - \frac{\kappa_0(\beta)}{2}f^2\right)\right], \tag{25}$$

which in turn yields

$$H_{\mathrm{MF}}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2}f^2 + \frac{1}{\beta}\log Z_X(\beta)\,\mathbb{I}. \tag{26}$$

Notably this correction to $H_{\mathrm{MF}}(\beta)$ is entirely *classical* [1]. This is hardly surprising, but emphasises that quantum effects *must* stem from non-commutativity. A truly quantum theory of thermodynamics therefore only makes meaningfully distinct predictions when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\mathrm{MF}}(\beta)$ possesses a closed form.

GENERATING-FUNCTION STRUCTURE OF THE INFLUENCE OPERATOR

Quenched propagator and Gaussian resummation

The starting point is the quenched propagator introduced in Sec. . In the imaginary-time interaction picture with respect to H_Q, the unnormalised reduced equilibrium operator takes the form

$$\begin{aligned}\bar{\rho}_Q(\beta) &= e^{-\beta H_Q}\,\langle W_\xi(\beta)\rangle_\xi,\\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\xi(\tau)\,\tilde{f}(\tau)\right],\end{aligned} \tag{27}$$

where $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ and the noise covariance is $\langle\xi(\tau)\xi(\tau')\rangle_\xi = K(\tau - \tau')$.

Because the bath is Gaussian, only the second cumulant of ξ is non-vanishing. Time-ordering introduces a non-trivial operator structure, but the key identity—proved in Appendix A—is that the Wick resummation of all pairings exponentiates exactly:

$$\langle W_\xi(\beta)\rangle_\xi = \exp\left[\frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau - \tau')\,\mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]\right]. \tag{28}$$

The proof is a direct application of Wick's theorem to the Dyson expansion of W_ξ: at order $2p$ in the expansion, there are $(2p)!/(2^p p!)$ distinct Wick pairings, each weighted by the two-point kernel $K(\tau_i - \tau_j)$. The prefactors from the Dyson expansion precisely cancel the pairing multiplicity, leaving only the singly-connected pairs.

We can exploit the symmetry $K(\tau - \tau') = K(\tau' - \tau)$ to fold the square domain onto the triangular one (see Appendix A), such that the exponent simplifies to

$$\Delta(\beta) \equiv \int_0^\beta d\tau\int_0^\tau d\tau'\, K(\tau - \tau')\,\tilde{f}(\tau)\tilde{f}(\tau'), \tag{29}$$

and the averaged reduced state takes the product form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q}\, e^{\Delta(\beta)}. \tag{30}$$

The influence exponent $\Delta(\beta)$ is quadratic in $\tilde{f}$ and depends on the bath exclusively through the two-point kernel K. Here the absence of higher-order bath objects is not an approximation, but an exact consequence of the Gaussian nature of the environment.

The ordered Green function and frequency compression

At this stage, the operator Δ is still formally defined as an integral over the bath degrees of freedom. There is however a particularly satisfying interpretation of this operator in which the effect of the bath is readily understood. To make this explicit, we work in the eigenbasis of H_Q. Let $H_Q|i\rangle = E_i|i\rangle$, so that the matrix elements of the interaction-picture operator are

$$\tilde{f}_{ij}(\tau) = \langle i|\tilde{f}(\tau)|j\rangle = f_{ij}\, e^{\omega_{ij}\tau}, \qquad \omega_{ij} \equiv E_i - E_j. \quad (31)$$

Substituting into Eq. (29) and expanding in the eigenbasis, the influence exponent takes the spectral form

$$\Delta(\beta) = \sum_{i,\ell} \Delta_{i\ell}\, |i\rangle\langle\ell|, \quad (32)$$

where evaluating the matrix elements of the operator product yields

$$\Delta_{i\ell} = \sum_j f_{ij}\, f_{j\ell}\, G^{>}(\omega_{ij}, \omega_{j\ell}), \quad (33)$$

where all bath and temperature dependence has been absorbed into the *ordered Green function*

$$G^{>}(\omega_1, \omega_2) \equiv \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau - \tau')\, e^{\omega_1\tau + \omega_2\tau'}. \quad (34)$$

In standard finite-temperature perturbation theory, analogous time-ordered integrals are evaluated using discrete Matsubara frequencies. Here, the integration instead projects the bath response directly onto the Laplace frequencies set by the system's own energy differences; the superscript $>$ records that $\tau > \tau'$ throughout the integration domain.

The double integral in Eq. (34) collapses to a one-dimensional Laplace transform upon changing variables to $u = \tau - \tau'$ and $v = \tau'$:

$$G^{>}(\omega_1, \omega_2) = \frac{e^{\beta\omega_1}\mathcal{K}(\omega_2) - \mathcal{K}(\omega_1)}{\omega_1 + \omega_2}, \quad (35)$$

where

$$\mathcal{K}(\omega) \equiv \int_0^\beta du\, K(u)\, e^{\omega u} \quad (36)$$

is the bare Laplace transform of the bath kernel. Eq. (35) is an exact closed form valid for all $\omega_1 + \omega_2 \neq 0$. Importantly, it implies that for any transition between levels in the reduced system, the entirety of the bath's influence is encoded by $\mathcal{K}$ evaluated at the transition frequency.

The resonant channel $\omega_1 + \omega_2 = 0$ requires separate treatment. Taking the limit carefully yields

$$R(\omega) \equiv G^{>}(\omega, -\omega) = \int_0^\beta (\beta - u)\, K(u)\, e^{\omega u}\, du, \quad (37)$$

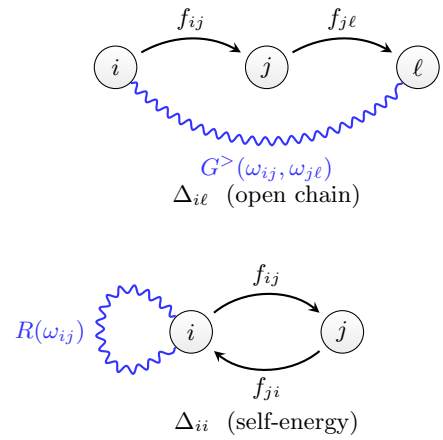

FIG. 1. Diagrammatic structure of the influence exponent $\Delta(\beta)$ in the H_Q eigenbasis. *Top*: The off-diagonal element $\Delta_{i\ell}$ (open chain) sums intermediate sites j, weighted by the bath propagator $G^{>}(\omega_{ij}, \omega_{j\ell})$ (wavy line) and two coupling insertions f (arrows). Energy is not conserved at the intermediate vertex. *Bottom*: The diagonal element Δ_{ii} (self-energy) is a closed loop $i \to j \to i$; the resonant condition $\omega_1 + \omega_2 = 0$ is satisfied, and the bath propagator collapses to $R(\omega_{ij})$.

where the factor $(\beta - u)$ is the available imaginary time after a bath correlation at separation u; the resonant kernel thus accumulates bath memory over the full thermal interval, regulated by the finiteness of β.

The structure of Eq. (33) admits a pleasingly simple diagrammatic reading, illustrated in Fig. 1. To isolate the physically distinct effects of the bath, we partition the influence exponent into its secular (diagonal) and mixing (off-diagonal) components, $\Delta = \Delta_\parallel + \Delta_\perp$.

The secular sector $\Delta_\parallel \equiv \sum_i \Delta_{ii}|i\rangle\langle i|$ singles out the *closed loops* $i \to j \to i$, for which $\omega_{ij} + \omega_{ji} = 0$ identically. The resonance condition forces the bath propagator onto $R(\omega_{ij})$, the object that accumulates bath memory over the whole thermal interval. The diagonal elements are thus the exact, non-perturbative *self-energy* dressing of the i-th energy level by all levels j it is coupled to:

$$\Delta_{ii} = \sum_j |f_{ij}|^2\, R(\omega_{ij}). \quad (38)$$

Conversely, the mixing sector $\Delta_\perp \equiv \sum_{i\neq\ell} \Delta_{i\ell}|i\rangle\langle\ell|$ describes *open chains* $i \to j \to \ell$, where energy is not conserved at the intermediate vertex. Using the closed form in Eq. (35), these components can be expressed directly in terms of the Laplace-transformed bath kernel $\mathcal{K}(\omega)$:

$$\Delta_{i\ell} = \frac{1}{\omega_{i\ell}} \sum_j f_{ij} f_{j\ell} \left[e^{\beta\omega_{ij}}\mathcal{K}(\omega_{j\ell}) - \mathcal{K}(\omega_{ij}) \right] \quad (i \neq \ell), \quad (39)$$

which encodes the coherent transitions between distinct energy levels mediated by the asymmetric fluctuations of the bath.

KMS symmetry and the symmetrised reduced state

The bath kernel $K(\tau)$ satisfies the KMS periodicity condition $K(\tau) = K(\beta-\tau)$, which in Laplace space reads

$$\mathcal{K}(-\omega) = e^{-\beta\omega}\,\mathcal{K}(\omega). \tag{40}$$

Substituting into the closed form Eq. (35) and simplifying gives a corresponding symmetry of the ordered Green function:

$$G^{>}(-\omega_2,-\omega_1) = e^{-\beta(\omega_1+\omega_2)}\,G^{>}(\omega_1,\omega_2). \tag{41}$$

Because transition frequencies telescope, $\omega_{ij}+\omega_{j\ell}=\omega_{i\ell}$, applying Eq. (41) to the matrix-element formula Eq. (33) immediately yields the *detailed-balance* constraint

$$\Delta_{\ell i} = e^{-\beta\omega_{i\ell}}\,\Delta^*_{i\ell}, \tag{42}$$

or in operator form $\Delta^\dagger = \Pi\,\Delta\,\Pi^{-1}$ where $\Pi \equiv e^{-\beta H_Q}$. Although Δ itself is not self-adjoint, this relation is a precise constraint: the product $\Pi\Delta$ is Hermitian.

This symmetry allows us to recast the unnormalised reduced state Eq. (30) into a manifestly positive, Hermitian form. Writing the bare Gibbs factor as $\Pi \equiv e^{-\beta H_Q}$ gives $\bar\rho_Q = \Pi\,e^{\Delta}$. By splitting $\Pi = \Pi^{1/2}\Pi^{1/2}$ and inserting the identity $\Pi^{-1/2}\Pi^{1/2} = \mathbf{1}$, we symmetrically distribute the bare weight:

$$\bar\rho_Q = \Pi^{1/2}\big(\Pi^{1/2}\,e^{\Delta}\,\Pi^{-1/2}\big)\Pi^{1/2}. \tag{43}$$

Applying the operator identity $A\,e^{B}A^{-1} = \exp(ABA^{-1})$ to the inner bracket yields

$$\bar\rho_Q = \Pi^{1/2}\,e^{S}\,\Pi^{1/2}, \qquad S \;\equiv\; \Pi^{1/2}\,\Delta\,\Pi^{-1/2}, \tag{44}$$

where S is the *symmetrised influence operator*. Its matrix elements evaluate to $S_{i\ell} = e^{-\beta\omega_{i\ell}/2}\Delta_{i\ell}$, which by Eq. (42) satisfy $S^*_{i\ell} = S_{\ell i}$. Thus, the modified exponent S is strictly self-adjoint. The partition function of the reduced system follows by cyclicity of the trace:

$$Z_Q \;\equiv\; \mathrm{Tr}[\bar\rho_Q] = \mathrm{Tr}[\Pi\,e^{S}] = \sum_i e^{-\beta E_i}\,(e^{S})_{ii}. \tag{45}$$

Since S is self-adjoint, $(e^S)_{ii} \geq e^{S_{ii}} = e^{\Delta_{ii}}$ by Jensen's inequality applied to the spectral weights $|\langle i|n\rangle|^2$, giving the bound

$$Z_Q \;\geq\; \sum_i e^{-\beta E_i+\Delta_{ii}}, \tag{46}$$

where the right-hand side is a Boltzmann sum with bath-shifted energies and equality holds when S is diagonal (i.e. when H_Q and Δ share an eigenbasis). The full result $Z_Q = \mathrm{Tr}[\Pi e^S]$ captures all level-mixing corrections through the off-diagonal elements of S.

THE HAMILTONIAN OF MEAN FORCE

Closure as an operator-algebra statement

With this, we may now give the general expression for the Hamiltonian of Mean-Force.

$$-\beta H_{\mathrm{MF}}(\beta) = \log\big(e^{-\beta H_Q}\,e^{\Delta(\beta)}\big). \tag{47}$$

If H_Q and Δ commuted, the logarithm would trivially give $-\beta H_Q + \Delta$. In general they do not, and the Baker–Campbell–Hausdorff (BCH) expansion

$$\log(e^{A}e^{B}) = A+\Phi(\mathrm{ad}_A)\,B+O(B^2), \qquad \Phi(x) \equiv \frac{x}{1-e^{-x}}, \tag{48}$$

generates a tower of commutators of $A = -\beta H_Q$ acting on $B = \Delta$. To understand the structure of these corrections, we invoke the partition of the influence exponent into its secular and mixing parts, $\Delta = \Delta_\parallel + \Delta_\perp$. The structural partitioning of these mean-force corrections is illustrated in Fig. 2. Because the secular piece $\Delta_\parallel$ strictly commutes with A, its adjoint action is annihilated ($\mathrm{ad}_A(\Delta_\parallel) = 0$).

Consequently, the generating function evaluated at $x = 0$ yields $\Phi(0) = 1$, allowing the parallel component to push straight through the linear BCH expansion:

$$-\beta H_{\mathrm{MF}} \approx \big(-\beta H_Q + \Delta_\parallel\big) + \Phi(\mathrm{ad}_{-\beta H_Q})\,\Delta_\perp + O(\Delta^2). \tag{49}$$

The secular zeroth-order shift $\Delta_\parallel$ thus groups directly with the bare system (reminiscent of the weak-coupling Lamb shift). Crucially however, while it acts geometrically like a classical energy shift, this term is still fundamentally *quantum* in origin because it is weighted by the bath Green functions, dynamically entangling the bare spectrum with the full spectral density of the environment. We can understand this directly by reference to Eq. (38). By defining a bath-dressed coupling operator $\tilde f$ with matrix elements $\tilde f_{ij} = f_{ij}\sqrt{R(\omega_{ij})}$, the secular exponent takes the exact quadratic form $\Delta_\parallel = (\tilde f\tilde f^\dagger)_{\mathrm{diag}}$. In the macroscopic limit where the system commutes with its coupling ($[H_Q, f] = 0$), the matrix f is entirely diagonal in the energy basis ($\omega_{ij} = 0$), and this dressed square collapses precisely to $\Delta_\parallel = \beta\lambda f^2$, where $R(0) = \beta\lambda$ is the explicit reorganisation energy defined in Eq. (23). The zeroth order BCH term therefore exactly recovers the classical commuting shift $H_{\mathrm{MF}} = H_Q - \lambda f^2$ of Eq. (26), and at zeroth order the non-commuting correction can be obtained by replacing $f \to \tilde f$. The remaining coherence-generating effects are then driven exclusively by the action of Φ on the perpendicular, off-diagonal open-chain fragments $\Delta_\perp$.

There is a saitsfyingly clean expression for these non-commuting corrections in the system energy eigenbasis. The off-diagonal basis operators $|i\rangle\langle j|$ are exact eigenoperators of the adjoint map: $\mathrm{ad}_{-\beta H_Q}|i\rangle\langle j| =$

$-\beta[H_Q, |i\rangle\langle j|] = -\beta\omega_{ij}|i\rangle\langle j|$. Consequently, the entire infinite tower of nested commutators generated by $\Phi(\mathrm{ad}_{-\beta H_Q})$ trivially evaluates to a simple scalar multiplication for each matrix element of the perpendicular fragment:

$$\Phi(\mathrm{ad}_{-\beta H_Q})\Delta_\perp = \sum_{i\neq j} \Phi(-\beta\omega_{ij})\,\Delta_{ij}|i\rangle\langle j|, \tag{50}$$

where $\Phi(x) = x/(1-e^{-x})$. The off-diagonal elements of the linear-in-Δ effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$.

This scalar evaluation exhibits a manifestly physical symmetry. If we reconstruct the effective matrix element using the symmetrised influence operator $S_{ij} = e^{-\beta\omega_{ij}/2}\Delta_{ij}$, the thermal asymmetry factor exactly absorbs the detailed-balance twist. The (off-diagonal) components on the effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$:

$$\begin{aligned}(H_{\mathrm{MF}})_{ij} &= -\frac{1}{\beta}\Phi(-\beta\omega_{ij})e^{\beta\omega_{ij}/2}S_{ij} \\ &= -\frac{\omega_{ij}/2}{\sinh(\beta\omega_{ij}/2)}\,e^{-\beta\omega_{ij}/2}\Delta_{ij}.\end{aligned} \tag{51}$$

Because $x/\sinh(x)$ is an explicitly even geometric function of the energy gap ω_{ij}, Eq. (51) guarantees that the linearly truncated effective Hamiltonian is perfectly Hermitian. Moreover, this exact thermal weighting reveals that bath-induced coherent mixing between levels is heavily exponentially suppressed for high-frequency transitions ($|\beta\omega_{ij}| \gg 1$), dynamically localising the leading mean force corrections to the thermodynamic energy shell.

This fortuitous cancellation reveals a deep structural connection between quantum thermodynamics and Lie algebra. The fundamental detailed balance condition of the bath imposes a thermal asymmetry $e^{-\beta\omega}$ on the bare transition amplitudes. Conversely, the BCH expansion—a purely abstract property of non-commuting logarithms—generates the parametric weighting $\Phi(-\beta\omega) \propto (1-e^{\beta\omega})^{-1}$. The fact that the Lie-algebraic geometry of the thermal exponential perfectly mirrors - and thereby absorbs - the thermodynamic detailed-balance twist of the environment is striking, and illustrative of the high degree of mathematical harmonisation underlying the mean-force formulation.

The foregoing parametric logic naturally generalises directly to all higher orders of the BCH expansion. Because the basis operators $|i\rangle\langle j|$ are strict eigenoperators of the adjoint map, the evaluation of any highly nested Lie commutator string spanning $O(\Delta^n)$ simply produces ordinary matrix products of the abstract fragments $\Delta_{ik_1}\Delta_{k_1k_2}\ldots\Delta_{k_{n-1}j}$, completely bypassing the need for functional derivatives. Over the energy eigenbasis, the entire non-linear algebraic sum reduces to evaluating n-step discrete pathways through the bare spectrum, where each composite transition amplitude is identically parametrically weighted by a universal, symmetric multivariate generalisation of Φ determined solely by Lie combinatorics.

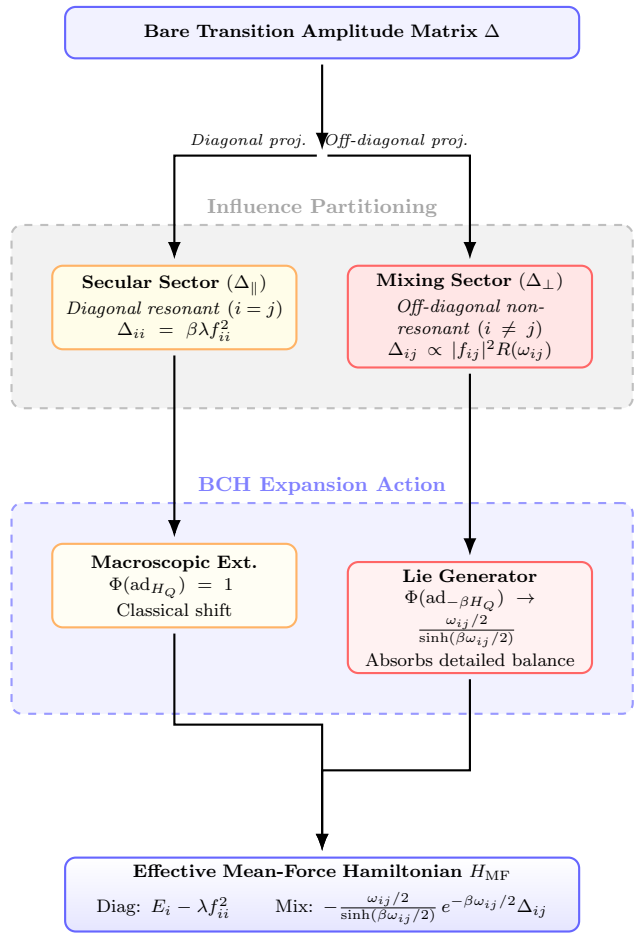

FIG. 2. Structural decomposition of the linear-order Mean-Force Hamiltonian. The parallel (secular) sector generates classical Lamb-like commuting phase shifts, entirely isolated from detailed balance effects. Concurrently, the perpendicular (mixing) sector passes through the Baker–Campbell–Hausdorff non-commutative generator, producing a geometric Lie scaling that perfectly absorbs the detailed balance thermal twist.

While this geometric pathing approach is exceptionally suggestive—and likely a numerically robust framework for resolving perturbative components—the abstract operator structure is more directly revealing of the core algebraic symmetries. Because the adjoint action obeys the rule $\mathrm{ad}_{H_Q}(f_n f_m) = f_{n+1}f_m + f_n f_{m+1}$ (where $f_n \equiv \mathrm{ad}_{H_Q}^n f$), every nested BCH commutator stays within the algebra generated by the adjoint chain of the coupling.

This ensures that even high-order nonlinear corrections introduce *no new bath objects*: the environment continues to enter exclusively through the fundamental spectral functions $\mathcal{K}(\omega)$ and $R(\omega)$, with coefficients determined solely by universal Lie combinatorics.

The closure content of $H_{\mathrm{MF}}(\beta)$ is therefore the following: the entire operator lives in the finite-dimensional algebra generated by $\{H_Q, \Delta\}$, with Δ quadratic in f and the bath entering exclusively through the two spectral objects $\mathcal{K}(\omega)$ and $R(\omega)$. Whether this algebra is itself finite-dimensional - and hence whether H_{MF} admits a closed form in a restricted operator family - depends on the structure of f and H_Q This statement leads to an important corollary. For continuous variable systems $H_Q = p^2/2m + V(x)$, if the potential $V(x)$ is a polynomial of degree greater than 2, the adjoint chain $\{f_n\}$ generally does not close in any finite-dimensional Lie algebra; instead, it generates the full universal enveloping algebra. Consequently, no finite representation of the mean force exists within the original operator class. This obstruction is analogous (in spirit) to the Groenewold–Van Hove no-go theorem on consistent quantisation of polynomial observables: the failure to close the Lie algebra under adjoint action corresponds to the impossibility of mapping the quantum evolution of non-quadratic potentials to a finite phase space flow. For such potentials, there is no *local operator* that exactly describes the reduced equilibrium state.

From a practical standpoint, the closure of the adjoint-chain is not necessary for an arbitrarily accurate approximation of $H_{\mathrm{MF}}(\beta)$, as the BCH logarithm $\log(e^{-\beta H_Q} e^{\Delta})$ still yields a controlled expansion. From this algebraic perspective, the closure problem offers a unifying view of strong-coupling approximations. Numerical schemes such as the polaron transformation [30, 56], hierarchical equations of motion [37], and chain-mapping/DMRG techniques [39] can be understood as distinct choices of a truncated operator family $\mathcal{A}$. The polaron frame corresponds to dressing $\mathcal{A}$ with coherent displacements, HEOM truncates the memory depth of the influence functional (dual to the operator degree in the adjoint chain), and chain mappings truncate the spatial extent of the harmonic bath. The 'closure' problem is thus equivalent to identifying a low-dimensional subalgebra that approximately contains the logarithm of the quenched propagator.

EXACT SOLUTION: GEOMETRIC REPRESENTATION

We now put the machinery of Secs. – to work in the transverse-coupling spin-boson model. The qubit is the prototypical rank-1 case of the general framework, yet its $\mathfrak{su}(2)$ algebra is rich enough to generate all the essential non-perturbative physics. As illustrated in Fig. 3, the exact qubit mean-force state $\mathbf{r}$ lives in a three-dimensional Bloch sphere, but is geometrically locked to the *coupling plane* (shaded) spanned by the system axis $\hat{\mathbf{n}}_s$ and the interaction axis $\hat{\mathbf{f}}$. The orientation of this plane is defined by the interaction phase ϕ_f, which directly determines the phase of the induced coherences in the reduced state.

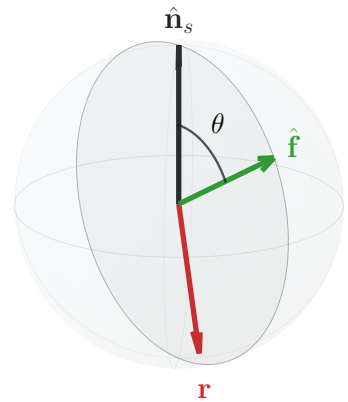

FIG. 3. 3D geometric overview of the exact qubit mean-force state. The system is defined by the bare axis $\hat{\mathbf{n}}_s$ and the interaction axis $\hat{\mathbf{f}}$ at polar angle θ and azimuthal angle ϕ_f. The exact reduced-state Bloch vector $\mathbf{r}$ is constrained to the shaded coupling plane. The interaction phase ϕ_f rotates the coupling plane and sets the phase of the steady-state coherences.

We first define our system Hamiltonian and coupling via

$$H_Q = \frac{\omega_q}{2} \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma}, \qquad f = \hat{\mathbf{f}} \cdot \boldsymbol{\sigma}, \tag{52}$$

where $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$ are unit vectors defining the system and coupling axes, respectively. The overall coupling strength g is omitted in the following derivation for clarity and can be re-introduced as a global scaling factor where needed. For the derivation it is convenient to work in the eigenbasis of H_Q (so $\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$). The final formulas are then rewritten covariantly in terms of $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}_\perp$.

The influence exponent may easily be calculated directly, using $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$. In the qubit energy basis ($\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$), we decompose the unit coupling vector as $\hat{\mathbf{f}} = (s\cos\phi_f, s\sin\phi_f, c)$, where $c \equiv \cos\theta$ and $s \equiv \sin\theta$ denote the longitudinal and transverse projections relative to the system axis, and ϕ_f is the azimuthal angle.

Writing

$$\begin{aligned} f &= c\sigma_z + f_-\sigma_+ + f_+\sigma_-, \\ f_\pm &\equiv s e^{\pm i\phi_f}, \end{aligned} \tag{53}$$

gives

$$\tilde{f}(\tau) = c\sigma_z + f_- e^{\omega_q \tau}\sigma_+ + f_+ e^{-\omega_q \tau}\sigma_-, \tag{54}$$

meaning $\Delta(\beta)$ is determined solely by the bath response kernels at the frequencies $0, \pm\omega_q$. Consequently, all bath dependence in Δ is compressed to scalar kernel data evaluated at the same frequencies. Using the notation of Sec. , the required bath objects are $\mathcal{K}(0)$, $\mathcal{K}(\pm\omega_q)$, and $R(0)$, $R(\pm\omega_q)$.

For a single qubit, the operator algebra closes almost trivially. Using $\sigma_+\sigma_- = (\mathbb{I} + \sigma_z)/2$, $\sigma_-\sigma_+ = (\mathbb{I} - \sigma_z)/2$, and $\sigma_\pm^2 = 0$, one finds

$$\begin{aligned} \tilde{f}(\tau)\tilde{f}(\tau') = &\left[c^2 + s^2\cosh\left(\omega_q(\tau-\tau')\right)\right]\mathbb{I} \\ &+ s^2\sinh\left(\omega_q(\tau-\tau')\right)\sigma_z \\ &+ c f_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+ \\ &+ c f_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-. \end{aligned} \tag{55}$$

Using the response kernels, the entirety of the bath's influence is compressed into two real, temperature-dependent response kernels: a longitudinal (resonant) kernel Σ_z and a transverse (non-resonant) kernel $\Sigma_\perp$, defined as

$$\Sigma_z \equiv \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \quad \Sigma_\perp \equiv \frac{4}{\omega_q}\left[\cosh\left(\frac{\beta\omega_q}{2}\right)\mathcal{K}(0) - e^{-\beta\omega_q/2}\mathcal{K}(\omega_q)\right]. \tag{56}$$

After performing the non-perturbative recombination (with the algebraic derivation detailed in Appendix C), the exact reduced density matrix ρ_Q takes the form:

$$\rho_Q = \frac{1}{2}\begin{pmatrix} 1 + \tanh\Theta & \kappa\,\mathrm{sech}\,\Theta e^{-i\phi_f} \\ \kappa\,\mathrm{sech}\,\Theta e^{i\phi_f} & 1 - \tanh\Theta \end{pmatrix}. \tag{57}$$

The operant parameters here — the renormalised thermal scale Θ and coherence strength κ — are given by

$$\Theta \equiv \mathrm{arctanh}(\gamma s^2\Sigma_z) - \frac{\beta\omega_q}{2}, \qquad \kappa \equiv \frac{-\gamma c s \Sigma_\perp}{\sqrt{1-(\gamma s^2\Sigma_z)^2}}, \tag{58}$$

where $\gamma = \tanh\chi/\chi$ is the non-perturbative resummation factor determined by the influence magnitude $\chi \equiv |\mathbf{h}_{\mathrm{eff}}|$. Vectorially, the traceless symmetrised influence is $M = \mathbf{h}_{\mathrm{eff}} \cdot \boldsymbol{\sigma}$ with

$$\mathbf{h}_{\mathrm{eff}} = \left(-\Sigma_\perp c s\cos\phi_f, -\Sigma_\perp c s\sin\phi_f, \Sigma_z s^2\right), \tag{59}$$

such that the crossover parameter factorises as

$$\chi = s\sqrt{s^2\Sigma_z^2 + c^2\Sigma_\perp^2} \equiv g^2\chi_0(\beta,\theta). \tag{60}$$

This representation reveals the reduced state ρ_Q as dependent on a renormalised thermal scale set by Θ, with its coherence dependent on the off-resonant bath response via κ.

From this matrix representation, we may directly identify the correspondence between the interaction geometry and the reduced state. The diagonal elements ρ_{00}, ρ_{11} determine the longitudinal population along $\hat{\mathbf{n}}_s$, while the off-diagonal elements ρ_{01}, ρ_{10} capture the coherences. Crucially, the phase of the coupling $e^{\pm i\phi_f}$ (the azimuthal orientation of $\hat{\mathbf{f}}$) is perfectly preserved in the state coherences, up to a sign $\mathrm{sgn}(\kappa)$. In terms of the Bloch vector $\mathbf{r} = (r_x, r_y, r_z)$, the steady-state is

$$r_z = \tanh\Theta, \qquad r_\perp = |\kappa\,\mathrm{sech}\,\Theta|, \qquad \phi_r = \phi_f + \pi\mathcal{H}(\Sigma_\perp c), \tag{61}$$

where $r_\perp$ denotes the transverse magnitude $\sqrt{r_x^2 + r_y^2}$. This confirms the intuition from Fig. 3: the state $\mathbf{r}$ always resides in the plane containing $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$. Any change in the coupling phase ϕ_f simply rotates this plane (and thus the coherence phase) around the system axis. Specifically, the relationship between interaction and coherence phase is manifest in the off-diagonal element:

$$\rho_{01} = \frac{\kappa}{2}\,\mathrm{sech}\,\Theta e^{-i\phi_f}, \tag{62}$$

which identifies ϕ_f as the generator of the steady-state coherence phase. The magnitude of the Bloch radius $r = \sqrt{r_z^2 + r_\perp^2}$ defines the purity of the state, while the tilt angle $\psi = \arctan(r_\perp/r_z)$ measures its geometric deflection from the bare thermal axis.

Weak- and strong-coupling limits

The weak- and strong-coupling limits are controlled by the single crossover variable χ. For $\chi \ll 1$,

$$\gamma(\chi) = 1 - \frac{\chi^2}{3} + O(\chi^4), \qquad \mathbf{u} = \mathbf{h}_{\mathrm{eff}} + O(\chi^3), \tag{63}$$

and the Bloch vector expands as

$$\mathbf{r} = -\tanh a\,\hat{\mathbf{n}}_s + \mathrm{sech}\,a\,\mathbf{h}_{\mathrm{eff},\perp} + \mathrm{sech}^2 a\,h_\parallel\,\hat{\mathbf{n}}_s + O(\chi^2). \tag{64}$$

Both the induced coherence ($\propto \mathbf{h}_{\mathrm{eff},\perp}$) and the longitudinal population correction ($\propto h_\parallel$) are $O(\chi^2)$.

For $\chi \gg 1$,

$$\gamma(\chi) \sim \chi^{-1}, \qquad \mathbf{u} = \tanh\chi\,\hat{\mathbf{h}}_{\mathrm{eff}} \to \hat{\mathbf{h}}_{\mathrm{eff}} + O(e^{-2\chi}), \tag{65}$$

so the dependence on the *magnitude* of $\mathbf{h}_{\mathrm{eff}}$ saturates and only its direction survives:

$$\mathbf{r} \to \frac{\hat{\mathbf{h}}_{\mathrm{eff},\perp} + \left(\hat{h}_\parallel\cosh a - \sinh a\right)\hat{\mathbf{n}}_s}{\cosh a - \hat{h}_\parallel\sinh a} \qquad (\chi \gg 1,\ a \text{ fixed}). \tag{66}$$

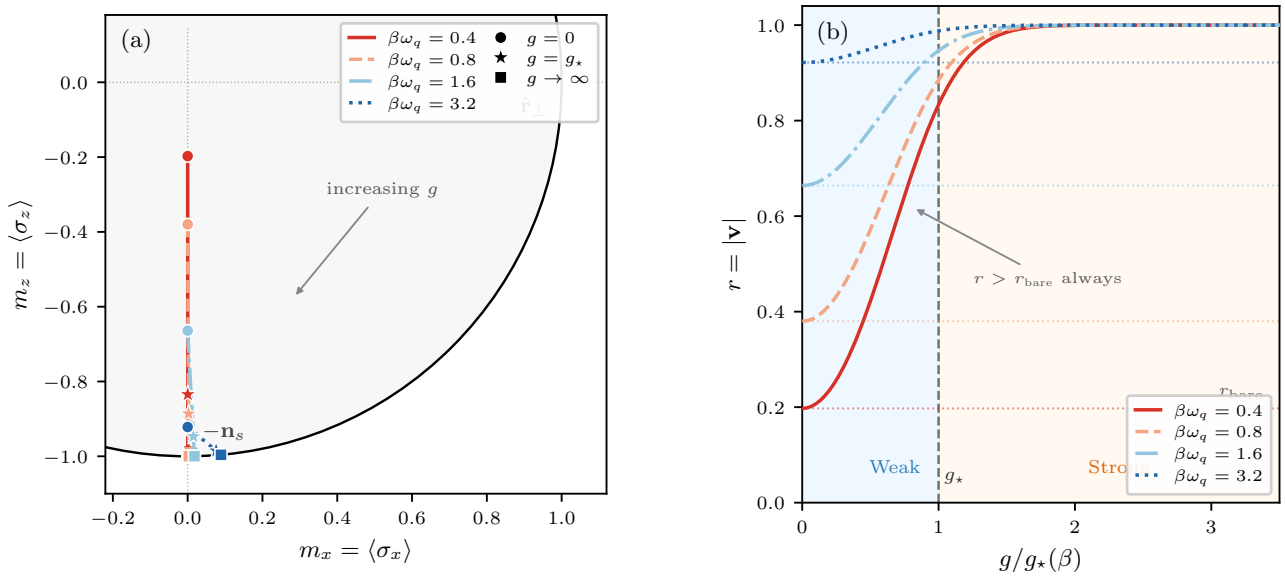

FIG. 4. Coupling-plane portrait of the qubit mean-force state. **(a)** Bloch disk cross-section in the coupling plane $(m_x, m_z) = (\langle\sigma_x\rangle, \langle\sigma_z\rangle)$ for $\theta = 3\pi/8$ (Ohmic bath, $\alpha = 1$, $\omega_c = 5\,\omega_q$). Each colored curve shows the trajectory of the exact mean-force Bloch vector $\mathbf{r}$ as χ increases from zero. Filled circles mark $\chi = 0$ (bare thermal state on the $-\hat{\mathbf{n}}_s$ axis); stars mark the crossover $\chi = 1$; squares mark the large-χ attractor. The gray arrow indicates the bare coupling direction $\hat{\mathbf{f}}$; the actual attractor lies in a different, temperature-dependent direction set by the bath-channel anisotropy (Sec.). **(b)** Bloch radius $r = |\mathbf{r}|$ vs. χ for the same temperatures. Dotted horizontals show the bare thermal purity $r_{\rm bare} = \tanh(\beta\omega_q/2)$. Every trajectory lies strictly above its own $r_{\rm bare}$ (Eq. (71)): bath coupling always increases purity regardless of temperature. The gain is largest at high temperature (red), where the bare state is near the maximally-mixed center of the disk.

The crossover is a smooth, exact interpolation controlled by χ: small χ gives linear response around the bare thermal axis $-\tanh a\,\hat{\mathbf{n}}_s$, while large χ locks the state to a temperature-dependent direction determined by the competition between $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{h}}_{\rm eff}$.

Relation to weak- and ultrastrong-coupling mean-force results

It is useful to make explicit how the exact qubit geometry derived above relates to the weak- and ultrastrong-coupling mean-force Gibbs (MFG) results in the literature, in particular the qubit limit discussed by Cresser and Anders. Their central ultrastrong statement is a *basis-selection* result: as the coupling scale $\lambda \to \infty$, the MFG state becomes diagonal in the eigenbasis of the coupling operator X, rather than in the bare Hamiltonian basis H_Q [25]. For a qubit with coupling axis $\hat{\mathbf{f}}$ and bare Hamiltonian axis $\hat{\mathbf{n}}_s$, this yields a limiting state whose Bloch vector points along $\hat{\mathbf{f}}$ (with a temperature-dependent magnitude set by the projected bare splitting).

Our exact Gaussian qubit solution is not in contradiction with that result, but it resolves a different structure. The qubit geometry obtained here proceeds in two distinct steps:

$$(\hat{\mathbf{n}}_s, \hat{\mathbf{f}}; \text{bath}) \longmapsto \mathbf{h}_{\rm eff} \longmapsto \mathbf{r}. \tag{67}$$

The first map constructs a *bath-dressed influence direction* through the symmetrised exponent, Eq. (59), which is explicitly anisotropic in the pair of directions $\{\hat{\mathbf{f}}_\perp, \hat{\mathbf{n}}_s\}$, with separate weights carried by the non-resonant and resonant bath channels. The second map then combines this dressed influence with the bare Gibbs factor Π to produce the final Bloch vector $\mathbf{r}$. The bath does not merely

rescale $\hat{\mathbf{f}}$; it remaps the coupling geometry relative to the system axis before thermal recombination.

Rewriting $\mathbf{h}_{\rm eff}$ in the coupling-adapted frame with $c \equiv \cos\theta$, $s \equiv \sin\theta$:

$$\mathbf{h}_{\rm eff} = s^2 c\,(\Sigma_z - \Sigma_\perp)\,\hat{\mathbf{f}} + s\,(\Sigma_\perp c + \Sigma_z s^2)\,\hat{\mathbf{n}}_\perp^{(f)}, \qquad (68)$$

where $\hat{\mathbf{n}}_\perp^{(f)} \equiv (\hat{\mathbf{n}}_s - c\hat{\mathbf{f}})/s$. The coupling-axis ultrastrong geometry of Cresser–Anders is recovered only when $\mathbf{h}_{\rm eff}$ aligns with $\hat{\mathbf{f}}$. From Eq. (68), this requires the resonant and non-resonant bath channels to satisfy a specific ratio that compensates for the coupling's geometric anisotropy:

$$\frac{\Sigma_z}{\Sigma_\perp} = -\frac{c}{s^2} \qquad (69)$$

(or equivalently $\Sigma_\perp c + \Sigma_z s^2 = 0$). which is a special condition on the bath parameters. Absent Eq. (69), the exact Gaussian qubit state may still enter the strong-coupling crossover (large χ), but approaches a bath-dressed attractor direction rather than the bare coupling axis itself (see the squares in Fig. 4a).

A useful summary is therefore:

> The Cresser–Anders ultrastrong theorem identifies the asymptotic *basis* selected in the singular coupling limit, while the present exact qubit geometry resolves the *route* to strong coupling in terms of a bath-anisotropic dressed influence vector built from resonant and non-resonant channels.

PHYSICAL REGIMES: RUNNING COUPLING, PURITY, AND TEMPERATURE DEPENDENCE

The exact formulas of Sec. contain three distinct physical phenomena that are absent from the bare-Gibbs or weak-coupling pictures. We examine each in turn.

The running coupling and temperature-driven crossover

From Eq. (60), the crossover parameter factorises as $\chi = g^2\chi_0(\beta)$, where the bath susceptibility $\chi_0(\beta)$ is the g-independent part. For an Ohmic bath $J(\omega) = \alpha\omega e^{-\omega/\omega_c}$, $\chi_0(\beta)$ is a monotone increasing function of β: the bath accumulates more imaginary-time dressing as the temperature decreases. This means that for any fixed coupling g, the dimensionless expansion parameter χ *grows* as the system cools, crossing through the non-perturbative scale $\chi \sim 1$ at the crossover temperature

$$\beta_\star(g) \;\text{ defined by }\; \chi_0(\beta_\star) = g^{-2}. \qquad (70)$$

The coupling g thus plays the role of a renormalization-group bare coupling, and $\chi(\beta, g)$ is the corresponding running coupling at scale β: the system flows from a weak-coupling fixed point at high temperature to a strong-coupling regime upon cooling.

Figure 5 illustrates this for three representative couplings. Panel (a) shows $\chi(\beta)$ rising through the crossover line $\chi = 1$ at $\beta_\star(g)$ (filled circles). Panels (b) and (c) show the corresponding evolution of the Bloch angle φ and Bloch radius r: both quantities track their bare values for $\beta < \beta_\star(g)$ (high temperature, weak coupling) and acquire significant strong-coupling corrections for $\beta > \beta_\star(g)$ (low temperature, strong coupling). The onset is smooth — it is controlled by the regular function $\gamma(\chi) = \tanh\chi/\chi$ — but the corrections accelerate steeply once $\chi \gg 1$.

Purity of the mean-force state

The purity of the qubit mean-force state is $\mathcal{P} = \frac{1}{2}(1 + r^2)$ with $r^2 = |\mathbf{r}|^2$. We can establish an exact, universal result:

> **Bath always purifies.** *For any positive spectral density $J(\omega) \geq 0$ and any coupling $g > 0$ and temperature $\beta > 0$,*
>
> $$r^2(g,\beta) \;>\; r^2_{\rm bare} \;=\; \tanh^2\!\left(\frac{\beta\omega_q}{2}\right), \qquad (71)$$
>
> *with equality only at $g = 0$.*

Proof sketch. The key ingredient is the sign of the longitudinal channel amplitude $\Delta_{z,0}$. Writing the imaginary-time bath kernel $K(u) \geq 0$, the channel amplitude is

$$\Delta_{z,0} = \frac{s^2}{\pi}\int_0^\beta (\beta - u)\,K(u)\,\sinh(\omega_q u)\,du. \qquad (72)$$

All three factors in the integrand are non-negative on $[0,\beta]$ (since $\beta - u \geq 0$, $\sinh(\omega_q u) \geq 0$), so $\Delta_{z,0} > 0$ and therefore $n_z = -\Delta_{z,0}/\chi_0 < 0$.

With $n_z < 0$, direct computation of $r^2 - \tanh^2 a$ from the exact Bloch vector formula gives the numerator

$$\mathcal{N} \;=\; \sinh^2\chi\big(n_z^2\sinh^2 a + \cosh^2 a\big) - n_z\,\sinh(2a)\,\tfrac{\sinh(2\chi)}{2} > 0, \qquad (73)$$

since both terms are positive when $n_z < 0$ and $a, \chi > 0$. The denominator is manifestly positive, so $r^2 > \tanh^2 a$. □

Limits. At $a \to 0$ (infinite temperature) the bare state is maximally mixed but the coupling gives

$$r\big|_{a=0} = \tanh\chi > 0, \qquad (74)$$

and at $\chi \to \infty$ (very strong coupling) one finds $r \to 1$ (pure state) regardless of temperature. Thus the mean-force state interpolates from the mixed bare state to a

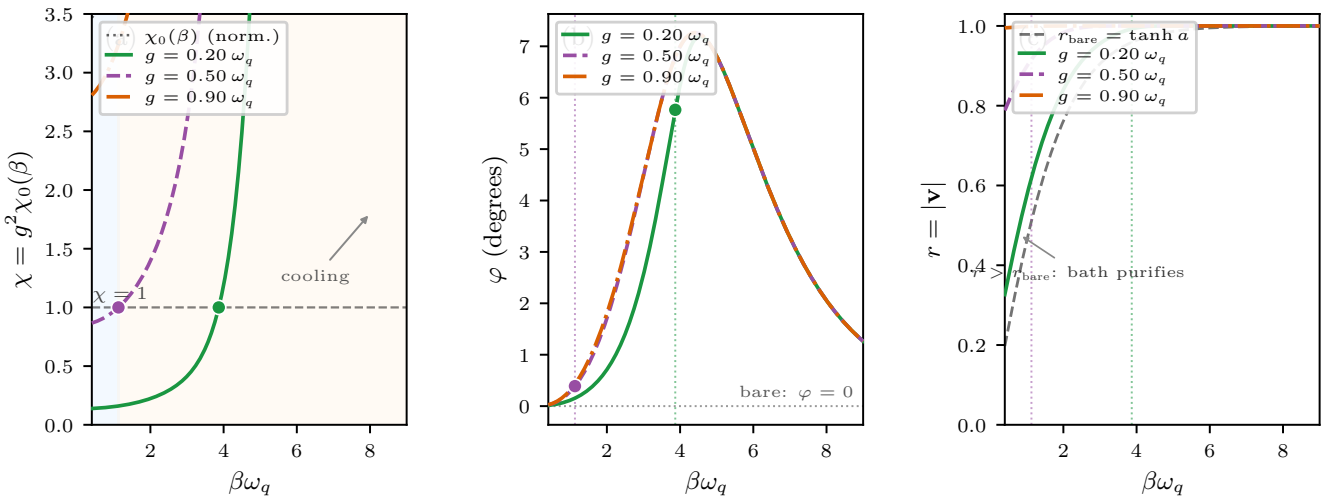

FIG. 5. Temperature-driven crossover for three coupling strengths ($g = 0.20, 0.50, 0.90\,\omega_q$; Ohmic bath, $\theta = \pi/6$). **(a)** Running coupling $\chi(\beta) = g^2\chi_0(\beta)$ versus $\beta\omega_q$. The dotted baseline shows the bath susceptibility $\chi_0(\beta)$ (same shape for all g, scaled by g^2). Filled circles mark the crossover temperature $\beta_\star(g)$ where $\chi = 1$. **(b)** Bloch angle $\varphi = \arctan(m_x, -m_z)$ (deviation from the bare thermal axis $-\hat{\mathbf{n}}_s$). The angle accumulates below $\beta_\star$ and saturates at the strong-coupling attractor direction. **(c)** Bloch radius $r = |\mathbf{r}|$, with the bare thermal purity $r_{\rm bare} = \tanh(\beta\omega_q/2)$ (black dashed) as a reference. For this Ohmic bath (bandwidth $2\omega_q$) the mean-force radius exceeds $r_{\rm bare}$ at all temperatures: the bath always purifies. The excess is largest at high temperature (β small, left of each $\beta_\star$ marker) where the bare state is nearly mixed, and smallest at low temperature where the bare state is already near-pure.

pure state as coupling increases, for any bath spectral function and any temperature.

The result is independent of the bandwidth or spectral shape of $J(\omega)$; widening the bandwidth increases $\chi_0(\beta)$ (driving the system to the strong-coupling regime at weaker g) but never reverses the purity inequality. Purity inversion would require $n_z > 0$, i.e. a bath kernel with negative imaginary-time correlations, which is impossible for any physical (positive) spectral density.

Purity landscape. Figure 6 displays the purity excess $\Delta r^2 = r^2 - \tanh^2 a > 0$ throughout the (g, β) plane for the Ohmic bath parameters used throughout. Panel (a) shows the absolute purity r^2; the white dashed contour is the crossover locus $\chi = 1$. Panel (b) shows Δr^2: the enhancement peaks at large g and high temperature (upper-right), where the bath lifts the near-maximally-mixed bare state far above its uncoupled value. At low temperature (bottom) the bare state is already near-pure and the excess vanishes.

Geometry of the state trajectory

The preceding analysis is unified by the Bloch disk portrait in Fig. 4. Several features of that figure deserve emphasis.

Temperature-dependent attractors. The squares in panel (a) (large-g attractors for each temperature) lie at *different* points on or near the unit circle boundary. This is because the direction $\hat{\mathbf{h}}_{\rm eff}$ depends on temperature through $b_{\rm nr}(\beta)$ and $b_{\rm r}(\beta)$: the bath-channel weights are frequency-resolved spectral integrals that change with β. By contrast, the Cresser–Anders ultrastrong limit predicts a *single* attractor direction aligned with $\hat{\mathbf{f}}$ (shown as the gray arrow) independent of temperature. The temperature-dependence of the exact attractor is the *temperature memory* of the mean-force state: even at arbitrarily strong coupling, the state encodes information about the temperature.

Purity along the trajectory. Panel (b) shows $r(g/g_\star)$ for each temperature. The dotted horizontal lines are $r_{\rm bare} = \tanh a$. By Eq. (71), every trajectory lies strictly above its own $r_{\rm bare}$ line for all $g > 0$. The effect is most visible at high temperature ($\beta\omega_q = 0.5$, red), where the bare state is near the maximally-mixed center and the bath-selected pure-state attractor offers a large purity gain. At low temperature ($\beta\omega_q = 6$, dark blue) the bare state is already near-pure and the excess is correspondingly small, though still nonzero. All trajectories converge to $r \to 1$ (pure state) as $g \to \infty$, consistent with the universal strong-coupling limit identified above.

Non-commutativity of limits. The limits $g \to \infty$ and $T \to 0$ do not commute. Taking $g \to \infty$ at fixed $T > 0$ locks the state to the bath-dressed attractor direction with finite-temperature purity. Taking $T \to 0$ at fixed g instead sends $a \to \infty$, and the state approaches the ground state $-\hat{\mathbf{n}}_s$ regardless of g. These two operations access qualitatively different corners of the state space.

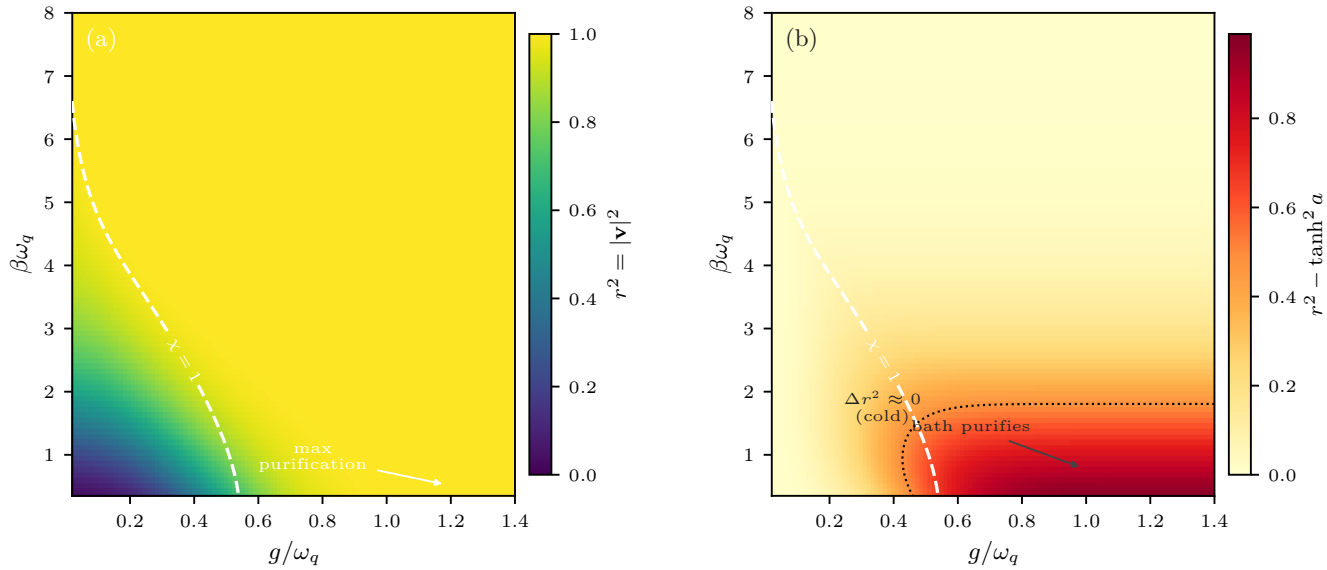

FIG. 6. Purity of the qubit mean-force state in the (g, β) plane ($\theta = \pi/6$, Ohmic bath). **(a)** Absolute purity $r^2 = |\mathbf{r}|^2$. The white dashed contour marks $\chi = 1$ (crossover locus). **(b)** Purity excess $\Delta r^2 = r^2 - \tanh^2 a > 0$ (sequential colour scale). By the exact result Eq. (71), coupling always increases purity for any harmonic bath; the maximum enhancement is at large g and high temperature, where the bath selects a preferred pure state from the near-maximally-mixed bare thermal ensemble.

NUMERICAL VALIDATION OF THE EXACT THEORY

Even in this simplest possible case, the physics of the reduced system is startlingly rich. The exact solution of the previous section therefore invites a detailed numerical analysis. With the exact form of the mean-force state, we can precisely characterise the behaviour of the system in different regimes of temperature and bath coupling.

First, to expose the coupling dependence cleanly, we reintroduce an explicit dimensionless strength g by writing the bath kernel as

$$K(u) = g^2 K_0(u). \tag{75}$$

All response kernels inherit the same quadratic scaling, such that

$$\Delta_z \to g^2 \Delta_z, \qquad \Sigma_\pm \to g^2 \Sigma_\pm, \qquad \chi \to g^2 \chi, \tag{76}$$

The entire non-perturbative recombination of bare Hamiltonian and influence is therefore controlled by the single dimensionless dressing parameter χ through

$$\gamma(\chi) = \frac{\tanh \chi}{\chi}. \tag{77}$$

This apparently innocuous expression encodes the entire non-perturbative recombination of bare Hamiltonian and influence, and functions as an *order* parameter for the crossover between weak and ultrastrong coupling regimes. The reduced equilibrium state transitions between these regimes as χ passes through unity,

$$\chi(\beta, g, \theta) \sim 1 \qquad \Longleftrightarrow \qquad g^2 \sim \chi_0(\beta, \theta)^{-1}. \tag{78}$$

This crossover condition $\chi(\beta, g, \theta) = 1$ therefore defines a coupling-dependent temperature scale

$$g_\star(\beta) = \chi_0(\beta, \theta)^{-1/2}, \tag{79}$$

and an equivalent crossover temperature $\beta_\star(g)$. No approximation has been made; $g_\star$ is an exact property of the solution.

The function $\gamma(\chi)$ is elementary but worth dwelling on. It is monotone decreasing from $\gamma(0) = 1$ to $\gamma(\infty) = 0$, with curvature that changes sign near $\chi \approx 1.2$. In the two limiting regimes:

$$\chi \ll 1: \quad \gamma \simeq 1 - \tfrac{\chi^2}{3}, \tag{80}$$

$$\chi \gg 1: \quad \gamma \simeq \chi^{-1} = (g^2 \chi_0)^{-1}. \tag{81}$$

These two expansions give a quantitative account of the two regimes identified qualitatively in Refs. [25, 26].

Weak-coupling regime, $\chi \ll 1$. When $\chi \ll 1$ (small g, or high temperature so that $\chi_0(\beta, \theta)$ is small), $\gamma \simeq 1$ and the density matrix Eq. (57) reduces to

$$\rho_Q \approx \frac{1}{\tilde{Z}} \begin{pmatrix} e^{-a}(1 + \Delta_z) & e^{-a}\sqrt{\chi^2 - \Delta_z^2} \\ e^{a}\sqrt{\chi^2 - \Delta_z^2} & e^{a}(1 - \Delta_z) \end{pmatrix} + O(\chi^3), \tag{82}$$

with $a = \beta\omega_q/2$. The off-diagonal entry—the bare-basis coherence—is $O(g^2)$, arising from $\Sigma_\pm \propto g^2$. The diagonal elements are similarly deformed from bare Gibbs at order g^2 through Δ_z. This is the perturbative dressing familiar from weak-coupling master equations, now derived exactly with no approximation on the spectral density.

Ultrastrong / low-temperature regime, $\chi \gg 1$. When $\chi \gg 1$ (large g, or very low temperature where χ_0 grows), $\tanh\chi \to 1$ and $\gamma \to \chi^{-1}$. The density matrix is then dominated by the normalised influence direction $\hat{M} = M/\chi$:

$$\rho_Q \approx \frac{e^{-\beta H_Q}\left(\mathbb{I} + \hat{M}\right)}{Z_Q}. \tag{83}$$

The state is sharply polarised along the axis defined by M, losing sensitivity to the bare Gibbs factor beyond what is needed to satisfy the KMS boundary conditions. The specific bath spectral weight matters only through the saturated invariants entering χ_0, not through any detailed frequency structure. In this limit the qubit effectively decouples from its own Hamiltonian and aligns with the interaction direction, the situation described as the "ultrastrong" mean-force fixed point in Refs. [25, 26].

The passage between these regimes is not a phase transition but a smooth crossover centered at $\chi \approx 1$. For the mixed coupling studied here, the crossover is marked by two distinct signatures in the system's differential response: a peak in the Bloch-angle susceptibility $\partial_g\varphi$ at $g_{\mathrm{peak},\varphi}$ and a peak in the Bloch-radius gradient $\partial_g r$ at $g_{\mathrm{peak},r}$. The location of the observable crossover signatures can be determined analytically by differentiating the exact Bloch-vector components with respect to g. Defining $a = \beta\omega_q/2$ and the dimensionless invariant $\chi = g^2\chi_0(\beta)$, the components m_x and m_z (fixing $\omega_q = 1$) are

$$m_x = \frac{\Sigma \tanh\chi}{\chi\cosh a - \Delta_z \sinh a \tanh\chi}, \tag{84}$$

$$m_z = \frac{\Delta_z \cosh a \tanh\chi - \chi \sinh a}{\chi\cosh a - \Delta_z \sinh a \tanh\chi}, \tag{85}$$

where $\Delta_z = g^2\Delta_{z0}$ and $\Sigma = g^2\Sigma_0$. The exact observable gradients are then given by

$$\partial_g\varphi = \frac{2g\Sigma_0\chi \sinh a \,\mathrm{sech}^2\chi}{\Sigma^2\tanh^2\chi + (\Delta_z\tanh\chi - \chi\tanh a)^2}, \tag{86}$$

$$\partial_g r = \frac{4g\chi_0 \tanh\chi\,\mathrm{sech}^2\chi}{r Z_Q^2}. \tag{87}$$

The exact location of these crossover signatures can be determined by setting the derivatives of the gradients to zero. Since the g-dependence enters solely through the invariant $\chi = (g/g_\star)^2$, we express the gradients as functions of χ and seek their maxima. In the high-temperature limit ($a \gg 1$), the condition $\frac{d}{dg}(\partial_g\varphi) = 0$ leads to the transcendental equation for the susceptibility peak:

$$1 - 4\chi\tanh\chi + \frac{4\chi c\,\mathrm{sech}^2\chi}{1 - c\tanh\chi} = 0, \tag{88}$$

where $c = \Delta_{z0}/\chi_0 \approx 0.7$ for the Ohmic bath at $\theta = \pi/4$. Solving this yields $\chi_{\mathrm{peak},\varphi} \approx 0.42$, pinning the peak at

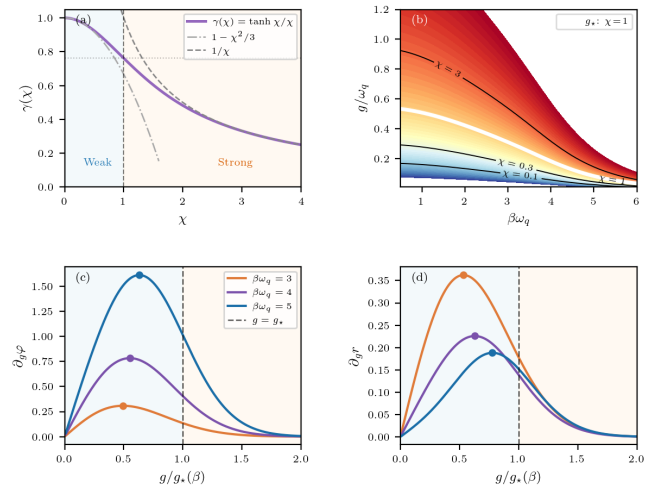

FIG. 7. Exact χ-crossover theory for the Ohmic bath (90) ($\theta = \pi/4$). *Panel (a)*: Universal crossover function $\gamma(\chi) = \tanh\chi/\chi$ between the weak-coupling ($1 - \chi^2/3$) and ultrastrong ($1/\chi$) limits. *Panel (b)*: Map of $\chi(\beta, g) = g^2\chi_0(\beta)$ showing the low-T divergence of the crossover scale. The white curve marks the exact center $g_\star(\beta) = \chi_0(\beta)^{-1/2}$. *Panels (c,d)*: Observable signatures of the crossover in the Bloch-angle susceptibility $\partial_g\varphi$ and Bloch-radius gradient $\partial_g r$ plotted against the scaled coupling $g/g_\star(\beta)$. Rescaling the coupling aligns the peak positions for all temperatures at the same horizontal markers. Shaded regions demarcate the weak (blue) and strong (orange) coupling regimes, with filled circles indicating the analytical peak locations.

$g_{\mathrm{peak},\varphi} \approx 0.65\, g_\star(\beta)$. Similarly, the purity gradient $\partial_g r$ achieves its maximum when

$$1 + 2\chi\left[\coth\chi - 3\tanh\chi + \frac{2c\,\mathrm{sech}^2\chi}{1 - c\tanh\chi}\right] = 0, \tag{89}$$

which is satisfied at $\chi_{\mathrm{peak},r} \approx 0.85$, or $g_{\mathrm{peak},r} \approx 0.92\, g_\star(\beta)$. Both conditions confirm that the observable signatures are universal multiples of the bath response scale $g_\star(\beta) = \chi_0^{-1/2}$. Indeed, by rescaling the coupling as $\xi = g/g_\star(\beta)$, both crossover signatures are pinned to fixed universal values of the invariant χ, aligning the horizontal peak positions for all temperatures (Fig. 7c,d). This demonstrates that the mean-force crossover is fundamentally governed by the accumulated bath response scale regardless of the environment statistics.

To make these predictions concrete we commit to a specific bath that permits exact analytic evaluation of all quantities. We choose a windowed Ohmic spectral density

$$J(\omega) = g^2\,\omega\, e^{-\omega/\omega_c}, \qquad \omega \in [0, 2\omega_q], \tag{90}$$

with cutoff $\omega_c = 5\omega_q$. The UV window $[0, 2\omega_q]$ is chosen to match typical numerical simulations where the bath state is truncated, while the exponential cutoff regularises the high-frequency tail [57]. Having specified the

bath, we further specialise simulations to $\theta = \pi/4$ and $\omega_q = 1$, as all other behaviour follows from the universal scaling. Figure 7 shows the exact χ-crossover theory for the Ohmic bath (90) ($\theta = \pi/4$). With these analytic predictions in hand, we now turn to comparison with numerical approximations.

NUMERICAL VALIDATION

The exact solution of the previous sections depends on the bath only through the kernel evaluated at three frequencies, $\omega \in \{0, \pm\omega_q\}$. To validate it we compare against Exact Diagonalisation (ED) of a truncated spin-boson model: the qubit is coupled to N_ω harmonic oscillators with frequencies $\omega_k = \omega_{\min} + k\,\Delta\omega$ and coupling amplitudes $c_k = \sqrt{J(\omega_k)\,\Delta\omega}$, each mode represented with a per-mode Fock cutoff $n_{\max}$.

A key feature of the comparison is that the analytic kernel integrals can be evaluated in two complementary ways: using the full continuum spectral density $J(\omega)$ (the *continuous analytic* branch) or restricted to the same N_ω discrete modes seen by ED (the *discrete analytic* branch). The discrete branch is the exact infinite-$n_{\max}$ limit of the ED simulation at fixed mode count, so discrepancies between ED and discrete analytic are purely numerical artefacts of finite Fock space. Discrepancies between the two analytic branches quantify how well the discrete mode set approximates the continuum.

Throughout this section we commit to the windowed Ohmic bath $J(\omega) = Q\,\tau_c\,\omega\,e^{-\tau_c\omega}$ with $Q = 10$, $\tau_c = 1$, window $[\omega_{\min}, \omega_{\max}] = [0.5, 8]\,\omega_q$, fixing $\omega_q = 2$, $\theta = \pi/4$, and introduce an explicit coupling prefactor g via $K(u) = g^2 K_0(u)$. The crossover parameter then reads $\chi(g, \beta) = g^2\chi_0(\beta)$ with $g_\star(\beta) = \chi_0(\beta)^{-1/2}$ the coupling scale at which the weak-to-ultrastrong crossover is centered.

One-dimensional benchmarks. Figure 8 shows the excited-state population p_{11} sweeping coupling g at fixed $\beta\omega_q = 2$, with $N_\omega = 40$ modes. At weak coupling ($\chi \ll 1$) all three curves — ED (green), discrete analytic (blue dashed), and continuous analytic (orange dotted) — agree within numerical precision. As the system crosses into the strong-coupling regime ($\chi \gtrsim 1$), the analytic branches diverge from ED. The crucial diagnostic is the relationship between the two analytic branches: the discrete branch correctly tracks the ultrastrong limit predicted by the continuum, while the ED simulation remains significantly elevated. This gap is not a failure of the theory, but a finite-Fock-space artefact: the polaron-like mean-force ground state requires mode displacements $\alpha_k \sim g c_k/\omega_k$, and representing it faithfully demands $n_{\max} \gtrsim |\alpha_k|^2$.

Figure 9 illustrates one consequence of this requirement. The cutoff sensitivity $\Delta p_{11} \equiv |p_{11}(n_{\max} = 4) - p_{11}(n_{\max} = 6)|$ grows with spectral bandwidth, with colder temperatures driving more pronounced sensitivity. This directly reflects the increased mode occupation at low frequencies needed to represent the displaced ultrastrong state.

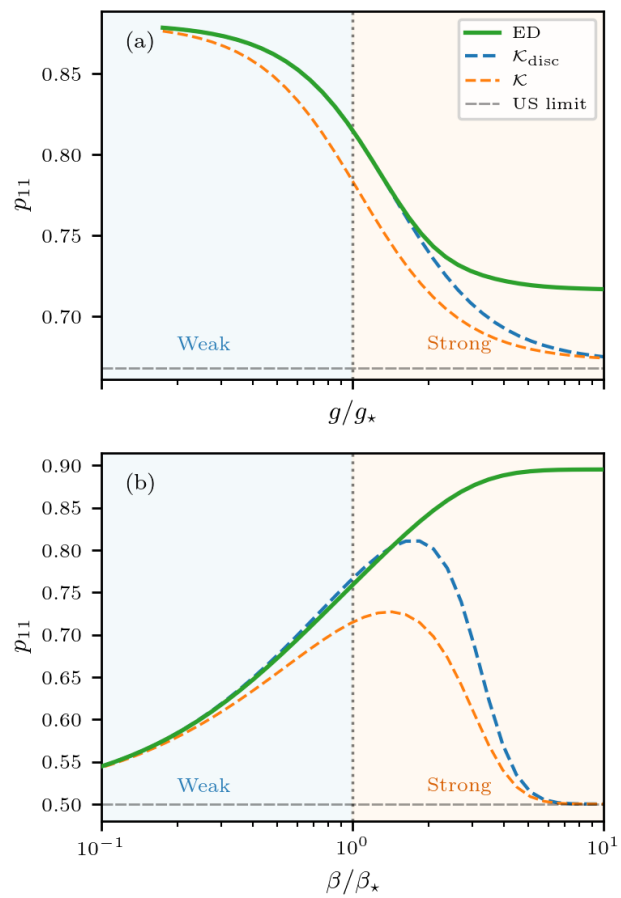

FIG. 8. Excited-state population p_{11} across the χ-crossover for $N_\omega = 40$ modes. Green solid: ED. Blue dashed: discrete analytic (same N_ω modes). Orange dotted: continuous analytic. Horizontal dashes mark the ultrastrong limit p_∞. The discrete analytic correctly reaches the US limit; the gap left by ED is a representability failure.

Two-dimensional landscape. Figures 8 and 9 trace one-dimensional slices through parameter space. To reveal the full structure we exploit GPU ED data covering the entire (g, β) plane at $\theta = \pi/4$. Figure 10 shows $N_\omega = 2$ modes (frequencies $0.5\omega_q$ and $8\omega_q$) with Fock cutoff $n_{\max} = 60$.

Panels (a) and (b) show the raw populations p_{00} from ED and from the discrete analytic formula respectively. Both heatmaps share the same broad topology: p_{00} decreases from the high-temperature Gibbs value ≈ 0.5 toward the ultrastrong limit as coupling and inverse temperature grow. The white dashed curve marks the $\chi_{\rm cont} = 1$ contour computed from the continuum Ohmic bath; this is exactly the $g_\star(\beta)$ crossover scale defined in Eq. (79). The most striking difference between the two panels is in the strong-coupling sector: the discrete an-

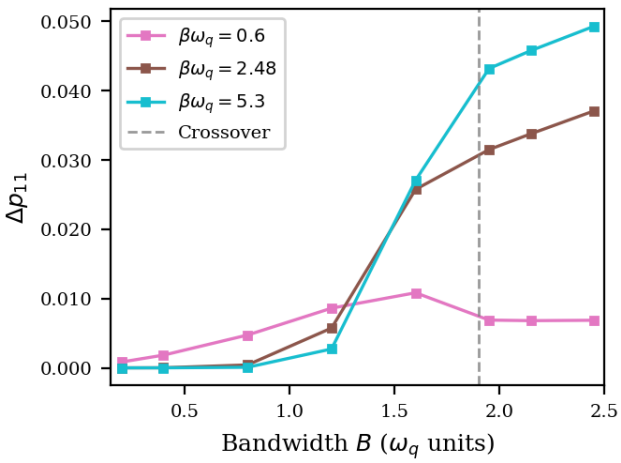

FIG. 9. Cutoff sensitivity Δp_{11} as a function of spectral bandwidth B (units of ω_q) at three inverse temperatures. Sensitivity grows with bandwidth and is larger at lower temperatures, reflecting the increased Fock-space requirement to represent low-frequency mode displacements as coupling grows.

alytic (panel b) predicts $p_{00} \to 0$ as the polaron ground state is reached, while the ED populations remain elevated (panel a) because the Fock space cannot accommodate the large mode displacements.

Panels (c) and (d) show the signed discrepancy $p_{00}^{\text{analytic}} - p_{00}^{\text{ED}}$ for the discrete and continuous branches respectively. Panel (c) reveals a systematic negative deviation: the two-mode discrete bath has coupling amplitude $c_1^2 = J(\omega_1)\Delta\omega \approx 22.7$ at $\omega_1 = 0.5\,\omega_q$ (with $\Delta\omega = 7.5\,\omega_q$ for $N_\omega = 2$), giving it far greater spectral weight than the continuum at the same g. Consequently, the discrete analytic predicts a more aggressive crossover than either the continuum or the ED simulation, and disc − ED is negative (analytic below ED) across the strong-coupling sector. Panel (d) is the physically transparent diagnostic: the continuum analytic uses the correct spectral density. Here the signed discrepancy is small and near zero throughout the entire $\chi_{\text{cont}} < 1$ half-plane, confirming that the HMF formula agrees with ED to within noise wherever the simulation is reliable. In the $\chi_{\text{cont}} > 1$ sector, the discrepancy grows and is again negative: the continuum analytic correctly predicts the lower-energy polaron state that ED cannot reach.

Representability bottleneck and convergence. The nature of the bottleneck is sharpened in Fig. 11. Panel (a) shows $p_{00}^{\text{ED}}(n_{\max})$ as a function of Fock cutoff for four representative $(g, \beta\omega_q)$ points spanning the coupling landscape, with the corresponding discrete analytic value plotted as a horizontal reference.

At weak coupling [blue, $g = 0.10$, $\beta\omega_q = 1.0$], ED reaches the analytic target within $n_{\max} \sim 10$: the polaron displacement is negligible and the standard Fock basis is entirely adequate. At moderate coupling [green, $g = 0.35$, $\beta\omega_q = 4.0$], convergence is achieved by $n_{\max} \approx 30$ with residual error below 0.5%.

At strong coupling the picture changes qualitatively. The orange curve [$g = 0.70$, $\beta\omega_q = 8.0$] displays *non-monotone* convergence: p_{00} increases with $n_{\max}$ over the accessible range $n_{\max} \in [6, 60]$, moving *away* from the analytic target. The physical origin is straightforward. The low-frequency mode at $\omega_1 = 0.5\,\omega_q$ carries spectral weight $c_1^2 \approx 22.7$, giving displacement $\alpha_1 = gc_1/\omega_1 \approx 9.5g$. True convergence requires $n_{\max} \gg \alpha_1^2 \approx 45$ at $g = 0.7$. For $n_{\max} \leq 60$ the Fock space supports only an incomplete polaron, and adding further Fock states moves the ED ground state energetically downward in a non-monotone fashion: the system partially accesses higher-occupation states before the true polaron minimum is reached. In the ultrastrong regime [red, $g = 1.36$, $\beta\omega_q = 16$], this effect is extreme: ED remains an order of magnitude above the analytic value across all $n_{\max}$ studied.

Panel (b) quantifies convergence by showing $|p_{00}^{\text{ED}}(n_{\max}) - p_{00}^{\text{disc.}}|$ on a log scale. The weak- and moderate-coupling points reach $< 0.5\%$ error well within accessible $n_{\max}$; the strong-coupling points show no systematic decrease within the studied range.

Panel (c) maps the $n_{\max} = 60$ residual $|p_{00}^{\text{ED}} - p_{00}^{\text{disc.}}|$ over the full (g, β) plane on a log scale, with the $\chi_{\text{cont}} = 1$ contour (white dashed) overlaid. The bottleneck is localized in the $\chi_{\text{cont}} \gtrsim 1$ band and grows monotonically with χ, while below $g_\star(\beta)$ the residual remains uniformly small. The $\chi = 1$ contour thus serves as an exact boundary for the regime where ED remains accurate: the HMF theory provides quantitative predictions on both sides, and the crossover parameter χ predicts precisely where numerical simulation fails.

Convergence to the Continuum Limit. To further isolate the representability bottleneck, Fig. 12 presents one-dimensional convergence slices sweeping both temperature and coupling strength. We plot three theoretical benchmarks: the exact, unrenormalized HMF theory (dotted gray), a "running coupling" heuristic (dashed black) that models the saturation of the finite-basis representation, and the asymptotic ultrastrong (US) limit $p_\infty = \cos^2(\theta/2)$ (dash-dotted gray).

In the thermal sweep at weak coupling (Panel a), the ED results for small local Hilbert spaces systematically approach the HMF theory as the number of bath modes and the Fock cutoff are increased. However, even at $g = 0.05$, a two-mode discretization with $n_{\max} = 6$ (red) displays visible deviations due to the discrete sampling of the spectral weight.

This effect is amplified in the coupling sweep (Panel b). While the raw HMF theory (dotted gray) continues toward the US limit as $\chi \to \infty$, the ED simulations for various discretizations diverge significantly once $\chi \gtrsim 1$. The running coupling heuristic (dashed black) successfully tracks the ED results into the strong-coupling regime by

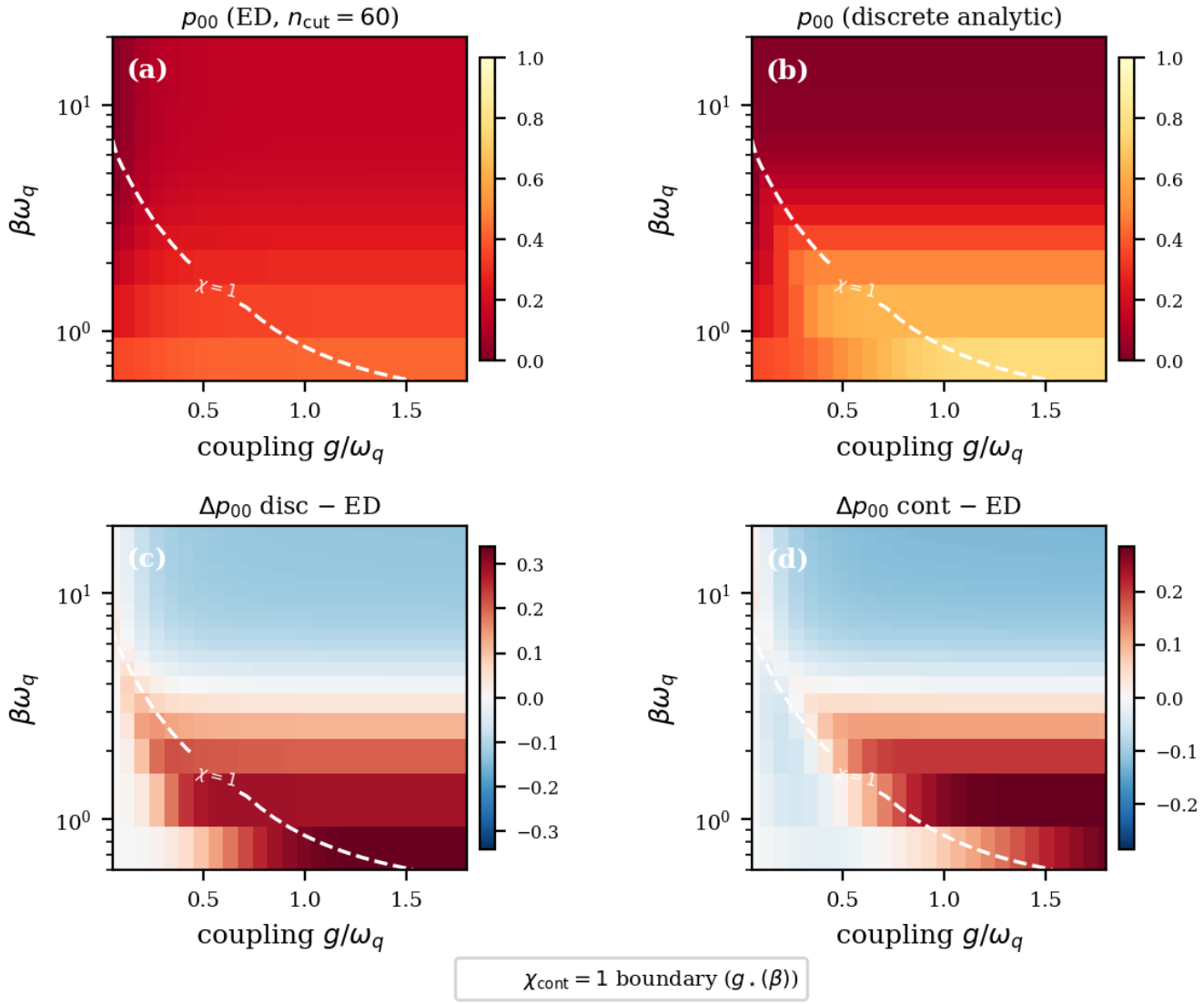

FIG. 10. Two-dimensional qubit-state landscape at $\theta = \pi/4$, $N_\omega = 2$ modes. *(a)* p_{00} from ED ($n_{\max} = 60$). *(b)* p_{00} from the discrete analytic formula (same two-mode bath, $n_{\max} \to \infty$). *(c)* Signed discrepancy Δp_{00} (discrete analytic − ED); the negative values throughout reflect the inflated spectral weight of the two-mode discretisation. *(d)* Same for the continuous analytic branch: the discrepancy is negligible in the $\chi_{\mathrm{cont}} < 1$ sector and grows only where the representability bottleneck prevents ED from reaching the polaron ground state. White dashed curve: $\chi_{\mathrm{cont}} = 1$ boundary ($\equiv g_\star(\beta)$, the exact crossover scale).

accounting for the representability bottleneck — the inability of the truncated Fock basis to support the large polaron displacements $\alpha_k \sim gc_k/\omega_k$. These benchmarks confirm that while the HMF theory is analytically exact, its numerical realization in standard Fock-space simulations requires careful accounting for basis truncation.

Summary. The numerical results establish three things simultaneously. First, the analytic HMF formula reproduces ED to within noise throughout the entire $\chi < 1$ sector, with no adjustable parameters. Second, the discrete analytic branch confirms that the theory is exact for arbitrarily small mode sets: the representability bottleneck is a property of the finite-dimensional simulation, not of the physical bath. Third, χ acts as both the physical crossover order parameter and the computational cost indicator: the minimum Fock cutoff required for ε-accurate ED scales as $n_{\max} \gtrsim (gc_k/\omega_k)^2$, which diverges as $g \to \infty$ at any fixed β. In the strong-coupling sector, where ED gives non-monotone and non-convergent results within accessible truncations, the exact HMF theory provides the only route to quantitative predictions.

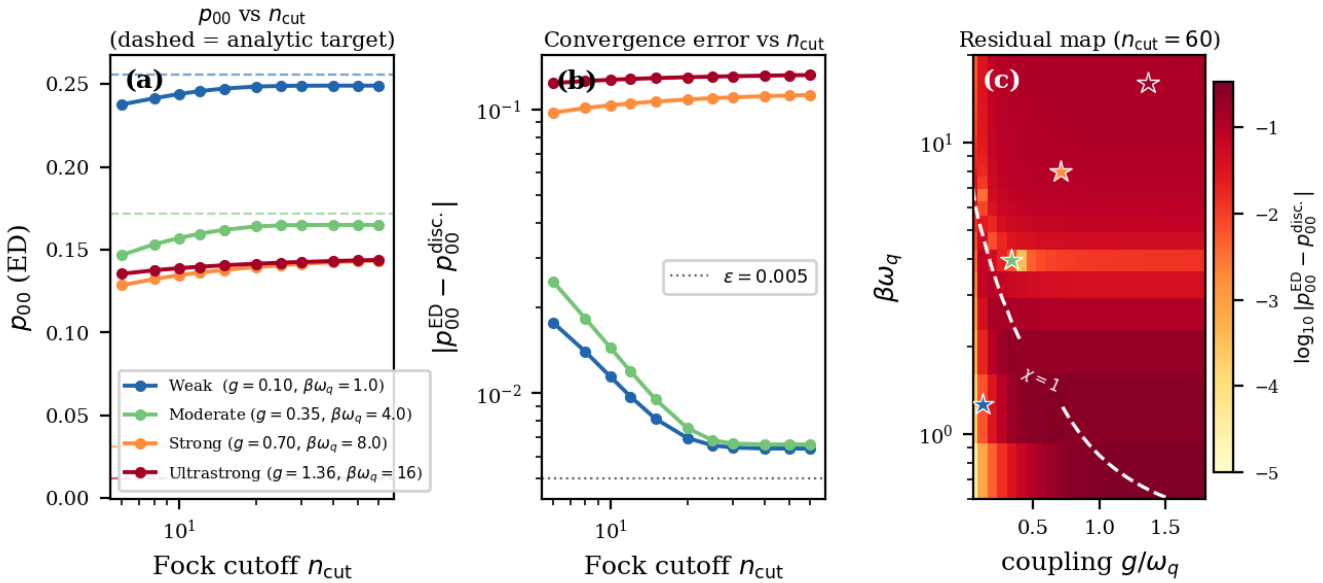

FIG. 11. Representability bottleneck at $\theta = \pi/4$, $N_\omega = 2$ modes. *(a)* $p_{00}^{\rm ED}(n_{\rm max})$ vs. Fock cutoff at four representative $(g, \beta\omega_q)$ points; dashed horizontals mark the discrete analytic target ($n_{\rm max} \to \infty$). *(b)* Log-scale convergence error $|p_{00}^{\rm ED} - p_{00}^{\rm disc.}|$ vs. $n_{\rm max}$. At strong coupling the curves display non-monotone convergence: p_{00} moves away from the analytic target within the accessible $n_{\rm max}$ range because the polaron displacement $\alpha_k \sim gc_k/\omega_k$ far exceeds $\sqrt{n_{\rm max}}$. *(c)* Two-dimensional residual map at $n_{\rm max} = 60$. Coloured stars mark the four representative points. The bottleneck is confined to $\chi_{\rm cont} \gtrsim 1$ with the exact crossover boundary (white dashed) partitioning the parameter space into a reliable weak-coupling sector and a Fock-space-limited strong-coupling sector.

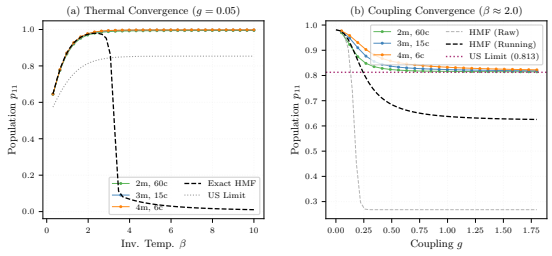

FIG. 12. Convergence of Exact Diagonalisation to the Mean-Force theory. *(a)* Temperature sweep at $g = 0.05$. ED results systematically approach the HMF theory as fidelity increases. *(b)* Coupling sweep at fixed β, showing how the running coupling heuristic (dashed black) captures the representability bottleneck of the truncated Fock basis, which prevents ED from reaching the raw HMF theory (dotted gray) or the exact US limit (dash-dotted gray).

DISCUSSION

This paper has three distinct but interlocking contributions: a structural criterion for operator representability of the HMF, an exact closed-form solution in the transverse-field qubit model, and a numerical characterisation of the crossover between the weak- and ultrastrong-coupling regimes. We discuss each in turn before drawing out the broader implications.

Closure and representability

The central algebraic result of Sec. is that the question "does a local closed-form HMF exist?" has a precise answer in terms of the operator algebra generated by repeated adjoint action of H_Q on the coupling f. Conditions (C1)–(C3) distil this into a hierarchy of closure requirements: adjoint closure of the coupling chain $\{f_n\}$, closure of that chain's quadratic span (which contains the Gaussian influence operator Δ), and BCH closure of commutators among the quadratic elements. When all three hold, $H_{\rm MF}(\beta)$ is a finite operator polynomial and can be extracted by a single BCH logarithm [48–50, 58]; when they fail, the HMF necessarily has support outside the target operator family and any representation within that family is a controlled approximation.

A notable consequence is the impossibility result for non-quadratic continuous-variable systems. For a Hamiltonian $H_Q = p^2/2m + V(x)$ with a polynomial potential $\deg V > 2$, the adjoint chain $\{f_n\}$ does not close in any finite-dimensional algebra: the chain grows without bound and generates the full universal enveloping algebra. This obstruction is analogous in spirit to the Groenewold–Van Hove theorem on consistent polynomial quantisation, and extends it to the equilibrium context:

for such potentials no finite local operator can represent the exact reduced equilibrium state.

On the constructive side, the BCH form Eq. (47) provides a systematic generator of the HMF in any regime. The separation of operator structure (carried entirely by the adjoint chain) from all bath and temperature dependence (entering only through the scalar kernel moments $C^{>}_{nm}(\beta)$) means that, in practice, one need only compute these moments — themselves one-dimensional Laplace-type integrals over the spectral density — and then perform matrix algebra in the operator space. This is computationally tractable even when the adjoint chain is long, and suggests a natural numerical scheme: truncate the chain at a finite depth, compute the corresponding moments, and use the BCH formula to assemble the approximate HMF. The approximation is then purely on the operator side; all bath physics enters exactly.

The exact qubit solution

The spin-boson model is the canonical testbed for open-quantum-system physics, yet the exact HMF in the transverse-coupling case has not, to our knowledge, been written down explicitly outside of weak-coupling or ultrastrong limiting forms [25, 26, 59]. The result (B11) achieves this using only three bath numbers: the kernel functional $\mathcal{K}$ evaluated at zero frequency, at the system resonance, and in the resonant channel $R(\omega_q)$.

The compression of the entire bath into three scalars is exact, not perturbative, and follows directly from the $\mathfrak{su}(2)$ closure of the adjoint chain. The algebra guarantees that no higher-frequency evaluations can appear: all imaginary-time convolutions collapse, via the ordered Green's kernel $\mathcal{G}^{>}(x, y)$ and its closed-form reduction (35), to a finite set of Laplace transforms evaluated on the system's own mass shell. This is the imaginary-time analogue of the fact that, in linear response, a two-level system couples to the bath only at its transition frequency.

The crossover parameter $\chi = \sqrt{\Delta_z^2 + \Sigma^2}$ that emerges from the $\mathfrak{su}(2)$ algebra is more than a convenient shorthand. Its square is the total accumulated bath response seen by the qubit in imaginary time — the analogue of the polaron displacement energy — and $\gamma(\chi) = \tanh\chi/\chi$ is the universal renormalisation factor that interpolates between the two fixed points. At $\chi \ll 1$, $\gamma \approx 1$ and the state is a weakly dressed bare Gibbs distribution. At $\chi \gg 1$, $\gamma \sim 1/\chi$ and the state is pinned to the interaction direction, reproducing the ultrastrong fixed point of Cresser and Anders [25, 26] exactly. The crossover between these regimes is smooth, parameter-free, and analytically located at $\chi = 1$ — the precise condition $g = g_\star(\beta)$.

This solution also sheds light on an apparent paradox that arises in the intermediate steps of the calculation. The influence operator $\Delta(\beta)$ has manifestly asymmetric ladder amplitudes, $\Sigma_+ \neq \Sigma_-$, which would seem to imply a non-Hermitian reduced state. The resolution is that the asymmetry is a gauge artefact of computing the influence before recombining with the bare Gibbs factor $e^{-\beta H_Q}$: the KMS relation Eq. (C9) precisely maps $\Sigma_+ = e^{\beta\omega_q}\Sigma_-$, and the product $e^{-\beta H_Q}e^{\Delta}$ is Hermitian by construction. The FDT is not an additional input but a consistency condition that is automatically enforced by the imaginary-time kernel.

The crossover as a representability bottleneck

The numerical results of Sec. add a dimension to the physical picture that is not visible from the analytic solution alone: the sensitivity of the simulated state to the representational limits of the Hilbert space.

The distinction between the discrete and continuous analytic branches in Fig. 8 is physically sharp. Both branches evaluate the same exact formula; crucially, the discrete-analytic branch (which uses the same mode set as ED) correctly tracks the continuum limit as $\chi \to \infty$. This proves that the HMF theory is exact for any mode set, and that a discrete bath possesses the same intrinsic ability to reach the ultrastrong limit as the continuum.

The divergence of ED from its own discrete analytic prediction as $\chi \gtrsim 1$ therefore reveals a **representability bottleneck**. In the strongly coupled regime, the bath oscillators acquire large polaron-like displacements. Representing such highly occupied states in a simulation with a fixed Fock cutoff $n_{\max}$ is fundamentally impossible; the truncated Hilbert space cannot accommodate the tails of the displaced ground state, forcing the simulation to stay at an artificially high energy level. This is a "mode occupation" failure, not a discretization failure.

The bandwidth analysis of Fig. 9 makes this quantitative. Choice of bandwidth B determines how the displacement "load" needed to reach the US limit is distributed among the available modes. The sharp peak in cutoff sensitivity is a direct marker of the available modes hitting their Fock boundaries as they attempt to polarize. This is not an arbitrary convergence artifact; it is the physical signature of trying to represent a highly non-local equilibrium state (in terms of mode occupancy) within a restricted representational space. The crossover parameter χ governs both the analytic structure of the state and the boundary where this representability bottleneck becomes insurmountable.

Connections to the literature

This work sits at the intersection of several lines of inquiry.

The Cresser-Anders framework [25, 26] gives the leading-order weak-coupling and exact ultrastrong-coupling forms for the qubit mean-force state. The present result interpolates between these limits exactly, identifies χ as the relevant dimensionless variable throughout, and gives the crossover scale $g_\star(\beta)$ in closed form.

The Du-Chen finite-bath HMF [27] generalises the mean-force concept to finite reservoirs by introducing a pair of effective Hamiltonians for system and bath. Our discrete-analytic branch is the commuting Gaussian instance of this construction: replacing the continuum by a discrete mode set is precisely the finite-bath limit, and the parametric sensitivity we observe in Fig. 9 is a concrete numerical realisation of the bath-feedback effects they discuss analytically.

The Burke-Iles-Smith structural study [28] analyses the operator content of the HMF through correlation-function methods. The potential-renormalisation perspective [19] provides a complementary bath-induced operator characterisation. Our closure criterion (C1)–(C3) provides a further algebraic characterisation: operator growth in the HMF corresponds precisely to the failure of one or more closure conditions.

More broadly, numerical methods that access $\bar{\rho}_Q(\beta)$ numerically — HEOM, stochastic Schrödinger/Liouville approaches [37, 40], chain mappings, and reaction-coordinate schemes [30–33] — are all, in a sense, performing the quenched-density average numerically. For a recent unifying overview connecting these approaches, see Ref. [39]. Our BCH/adjoint-chain organisation shows how to extract a closed-form operator representation from such an average whenever the algebra closes, and identifies the minimal set of bath evaluations required (three numbers in the qubit case). This suggests a hybrid programme: use numerical methods to compute kernel moments on a target spectral density, then assemble the HMF analytically via the BCH formula. Surveys of the growing thermodynamic strong-coupling programme can be found in Refs. [6, 7].

Limitations and outlook

The central analytical restriction of the present work is Gaussianity. The quenched representation and the cumulant expansion hold for any bath, but the exact collapse of Δ to a bilinear form in the adjoint chain relies on the bath cumulant hierarchy truncating at second order. Beyond the Gaussian sector, higher cumulants $K^{(n)}$ with $n \geq 3$ enter (10) and introduce products $f_{n_1} \cdots f_{n_k}$ of arbitrarily high order. Closure of such a family in a finite algebra is generically much harder. The non-Gaussian extension therefore represents the natural next step, and the adjoint-chain organisation already provides the skeleton: each additional cumulant order contributes a new layer of kernel moments multiplying operator monomials of corresponding degree, and the BCH structure persists.

Within the Gaussian sector, the qubit model studied here is the simplest closed algebra. The next natural target is multi-qubit systems, where the adjoint chain generated by H_Q will generically involve k-body Pauli strings. Closure then requires the target family to contain those strings, and the kernel-moment computation becomes a matrix problem in an enlarged operator space. Whether the HMF remains a two-body operator for a chain of qubits, or generates genuinely k-local terms, is a question our framework can in principle answer.

Finally, the crossover physics identified here has immediate consequences for the growing experimental literature on strong-coupling thermodynamics with superconducting circuits and semiconductor quantum dots [6, 7]. The crossover scale $g_\star(\beta) = \chi_0(\beta)^{-1/2}$ translates directly into a device-parameter threshold: below it, the bare Gibbs description is adequate; above it, the mean-force correction is non-perturbative. The quantitative agreement between the exact formula and ED simulation in the weak-coupling sector provides a parameter-free benchmark that can be used to validate strong-coupling simulation protocols in those settings.

Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\tilde{f}(\tau)\right], \tag{A1}$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau)\xi(\tau')\rangle = K(\tau-\tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau') \\ \times \mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]\Bigg].\end{aligned} \tag{A2}$$

The Dyson expansion of the time-ordered exponential is given by

$$\begin{aligned}W_\xi(\beta) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\cdots d\tau_n \\ \times \mathcal{T}_\tau\left[\xi(\tau_1)\tilde{f}(\tau_1)\cdots\xi(\tau_n)\tilde{f}(\tau_n)\right].\end{aligned} \tag{A3}$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors.

Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau\left[\xi_1\tilde{f}_1\cdots\xi_n\tilde{f}_n\right] = (\xi_1\cdots\xi_n)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_n\right], \tag{A4}$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi\rangle = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\,\langle\xi_1\cdots\xi_{2p}\rangle\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A5}$$

By Wick's theorem, the even moments are given by the sum over all pairings P:

$$\langle\xi_1\cdots\xi_{2p}\rangle = \sum_P\prod_{(i,j)\in P}K(\tau_i-\tau_j). \tag{A6}$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\,K(\tau-\tau')\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]. \tag{A7}$$

The p-th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables (τ_{2r-1},τ_{2r}), we write

$$X^p = \left(\frac{1}{2}\right)^p\int d^{2p}\tau\prod_{r=1}^{p}\left[K_{(2r-1)(2r)}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right], \tag{A8}$$

so that the full series is

$$\begin{aligned} e^X = \sum_{p=0}^{\infty}\frac{1}{p!}\left(\frac{1}{2}\right)^p\int d^{2p}\tau\left[\prod_{r=1}^{p}K_{(2r-1)(2r)}\right] \\ \times\left[\prod_{r=1}^{p}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right]. \end{aligned} \tag{A9}$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0,\beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau[\tilde{f}_1\cdots\tilde{f}_{2p}] = \tilde{f}_{\pi(1)}\cdots\tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau[\tilde{f}_{2r-1}\tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^{p}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right] = \mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A10}$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1,\ldots,p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P, there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\left(\sum_P\prod_{(i,j)\in P}K_{ij}\right)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A11}$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\begin{aligned} \langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau' K(\tau-\tau') \\ \times\mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]\Bigg]. \end{aligned} \tag{A12}$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\,K(\tau-\tau')\,\mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]. \tag{A13}$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right] = \Theta(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau')+\Theta(\tau'-\tau)\tilde{f}(\tau')\tilde{f}(\tau). \tag{A14}$$

Substituting this into the integral and splitting the domain yields

$$\begin{aligned} \Delta(\beta) = \frac{1}{2}\iint_{\tau>\tau'}d\tau d\tau'\,K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') \\ +\frac{1}{2}\iint_{\tau'>\tau}d\tau d\tau'\,K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau). \end{aligned} \tag{A15}$$

In the second term, we perform the swap of dummy variables $(\tau,\tau')\mapsto(\tau',\tau)$. The region $\tau'>\tau$ becomes $\tau>\tau'$, the operator product transforms to $\tilde{f}(\tau)\tilde{f}(\tau')$, and the kernel becomes $K(\tau'-\tau)$. Thus,

$$\begin{aligned} \frac{1}{2}\iint_{\tau'>\tau}d\tau d\tau'\,K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau) \\ = \frac{1}{2}\iint_{\tau>\tau'}d\tau d\tau'\,K(\tau'-\tau)\tilde{f}(\tau)\tilde{f}(\tau'). \end{aligned} \tag{A16}$$

Using the symmetry of the equilibrium kernel, $K(\tau - \tau') = K(\tau' - \tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau - \tau')\tilde{f}(\tau)\tilde{f}(\tau'). \tag{A17}$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2}\sigma_z, \qquad f = c\,\sigma_z - s\,\sigma_x, \tag{B1}$$

with $c = \cos\theta$, $s = \sin\theta$. The adjoint chain $f_n \equiv \mathrm{ad}^n_{H_Q}(f)$ is given by

$$f_n = \mathbf{f}_n \cdot \boldsymbol{\sigma}, \qquad \mathbf{f}_n = \begin{cases} (-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1,\ n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1,\ n \text{ odd}. \end{cases} \tag{B2}$$

The recurrence ensures that $f_{n\geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z-component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C^{>}_{nm} f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbb{I} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n>m} A_{nm}\,(\mathbf{f}_n \times \mathbf{f}_m)\cdot\boldsymbol{\sigma}, \tag{B3}$$

up to a scalar shift $\Delta_0\mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{f}_n \times \mathbf{f}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy-plane.
- For $m = 0$: $\mathbf{f}_n \times \mathbf{f}_0$ generates terms in the xy-plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0\mathbb{I} + \Delta_x\sigma_x + \Delta_y\sigma_y + \Delta_z\sigma_z, \tag{B4}$$

where the coefficients are polylogarithmic series in ω_q:

$$\Delta_x = -2cs \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\,\omega_q^{2\ell+1}, \tag{B5}$$

$$\Delta_y = -2ics \sum_{\ell\geq 1} A_{2\ell,0}(\beta)\,\omega_q^{2\ell}, \tag{B6}$$

$$\Delta_z = -2s^2 \sum_{k\geq 1,\ell\geq 0} A_{2k,2\ell+1}(\beta)\,\omega_q^{2k+2\ell+1} - 2s^2 \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\,\omega_q^{2\ell+1}. \tag{B7}$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^{>}(0, \pm\omega_q)$ and $\mathcal{G}^{>}(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\mathrm{MF}} = \log(e^{-\beta H_Q}e^{\Delta})$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0\mathbb{I} = \boldsymbol{\Delta}\cdot\boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\sigma})^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\Delta})\mathbb{I} \equiv \chi^2\mathbb{I}, \tag{B8}$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+\Sigma_-$ in the ladder gauge). This "generalized nilpotency" condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^{\Delta} = e^{\Delta_0}\left[\cosh\chi\,\mathbb{I} + \frac{\sinh\chi}{\chi}\,M\right]. \tag{B9}$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\mathrm{SL}(2,\mathbb{C})$ matrices:

$$\bar{\rho}_Q = e^{-\beta H_Q}e^{\Delta} = e^{\Delta_0}\begin{pmatrix} e^{-\beta\omega_q/2} & 0 \\ 0 & e^{\beta\omega_q/2}\end{pmatrix}\begin{pmatrix} \cosh\chi + \varphi\Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh\chi - \varphi\Delta_z\end{pmatrix}, \tag{B10}$$

with $\varphi = \sinh\chi/\chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r}\cdot\boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\mathrm{MF}} = c_0\mathbb{I} - \frac{1}{\beta}\frac{\operatorname{arctanh} r}{r}\,\mathbf{r}\cdot\boldsymbol{\sigma}. \tag{B11}$$

This formula represents the analytic summation of the entire BCH commutator tower. The "closure" discussed in Sec. manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

Appendix C: Exact Qubit Solution in the Ladder Basis

We provide the manual derivation of the exact qubit mean-force state using the ladder basis $\sigma_\pm$. Starting from the interaction-picture operator product

$$\begin{aligned}\tilde{f}(\tau)\tilde{f}(\tau') = &\Big[c^2 + s^2\cosh\big(\omega_q(\tau-\tau')\big)\Big]\mathbb{I}\\ &+ s^2\sinh\big(\omega_q(\tau-\tau')\big)\sigma_z\\ &+ cf_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+\\ &+ cf_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-,\end{aligned} \tag{C1}$$

the influence exponent $\Delta = \int_0^\beta d\tau\int_0^\tau d\tau' K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau')$ evaluates to

$$\Delta = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_+\sigma_+ + \Delta_-\sigma_-, \tag{C2}$$

with coefficients

$$\Delta_z = s^2\Sigma_z, \tag{C3}$$

$$\Delta_\pm = -2cf_\mp\Sigma_\pm, \tag{C4}$$

where $f_\pm = se^{\pm i\phi_f}$ and the primitive integrals are

$$\Sigma_z = \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \tag{C5}$$

$$\Sigma_\pm = \frac{1}{\omega_q}\left[\left(1 + e^{\pm\beta\omega_q}\right)\mathcal{K}(0) - 2\mathcal{K}(\pm\omega_q)\right]. \tag{C6}$$

The unnormalised state is $\bar{\rho}_Q = e^{-(\beta\omega_q/2)\sigma_z}e^{\Delta}$. To find the normalized state and its Bloch vector, we perform a symmetric decomposition

$$\bar{\rho}_Q = \Pi^{1/2}e^{S}\Pi^{1/2}, \qquad S = \Pi^{1/2}\Delta\Pi^{-1/2} \tag{C7}$$

where $\Pi = e^{-\beta\omega_q\sigma_z/2}$. Using $\Pi^{1/2}\sigma_\pm\Pi^{-1/2} = e^{\mp\beta\omega_q/2}\sigma_\pm$, the symmetric influence operator S becomes

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + (\Delta_+e^{-\beta\omega_q/2}\sigma_+ + \Delta_-e^{\beta\omega_q/2}\sigma_-). \tag{C8}$$

Substituting the expressions for $\Delta_\pm$ and using the KMS relation

$$\Sigma_- = e^{-\beta\omega_q}\Sigma_+, \tag{C9}$$

we define the real symmetrised transverse response

$$\Sigma_\perp \equiv 2e^{-\beta\omega_q/2}\Sigma_+ = 2e^{\beta\omega_q/2}\Sigma_-. \tag{C10}$$

The transverse coefficients then simplify to

$$\Delta_+e^{-\beta\omega_q/2} = -cse^{-i\phi_f}(2e^{-\beta\omega_q/2}\Sigma_+) = -cse^{-i\phi_f}\Sigma_\perp, \tag{C11}$$

with $\Delta_\perp \equiv -cs\Sigma_\perp$. Recombining the ladder operators into Cartesian form, we obtain

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_\perp(\sigma_x\cos\phi_f + \sigma_y\sin\phi_f), \tag{C12}$$

which is the generator of a rotation around an axis in the coupling plane. The influence magnitude $\chi = \sqrt{\Delta_z^2 + \Delta_\perp^2}$ determines the exponential $e^S = \cosh\chi(\mathbb{I} + \gamma M)$ with $\gamma = \tanh\chi/\chi$ and $M = S - \Delta_0\mathbb{I}$. The final state $\rho_Q = Z_Q^{-1}\Pi^{1/2}e^S\Pi^{1/2}$ follows by direct matrix multiplication, yielding Eq. (57) in the main text.

[1] G. McCaul, Mean-force hamiltonians from influence functionals, arXiv preprint arXiv:2602.13146 (2026), arXiv:2602.13146 [quant-ph].

[2] M. Campisi, P. Talkner, and P. Hänggi, Fluctuation Theorem for Arbitrary Open Quantum Systems, Physical Review Letters **102**, 210401 (2009).

[3] P. Talkner and P. Hänggi, *Colloquium* : Statistical mechanics and thermodynamics at strong coupling: Quantum and classical, Reviews of Modern Physics **92**, 041002 (2020).

[4] A. S. Trushechkin, M. Merkli, J. D. Cresser, and J. Anders, Open quantum system dynamics and the mean force Gibbs state, AVS Quantum Science **4**, 012301 (2022).

[5] U. Seifert, First and Second Law of Thermodynamics at Strong Coupling, Physical Review Letters **116**, 020601 (2016).

[6] S. Campbell, I. D'Amico, M. A. Ciampini, J. Anders, N. Ares, S. Artini, A. Auffèves, L. Bassman Oftelie, L. P. Bettmann, M. V. S. Bonança, T. Busch, M. Campisi, M. F. Cavalcante, L. A. Correa, E. Cuestas, C. B. Dag, S. Dago, S. Deffner, A. Del Campo, A. Deutschmann-Olek, S. Donadi, E. Doucet, C. Elouard, K. Ensslin, P. Erker, N. Fabbri, F. Fedele, G. Fiusa, T. Fogarty, J. Folk, G. Guarnieri, A. S. Hegde, S. Hernández-Gómez, C.-K. Hu, F. Iemini, B. Karimi, N. Kiesel, G. T. Landi, A. Lasek, S. Lemziakov, G. Lo Monaco, E. Lutz, D. Lvov, O. Maillet, M. Mehboudi, T. M. Mendonça, H. J. D. Miller, A. K. Mitchell, M. T. Mitchison, V. Mukherjee, M. Paternostro, J. Pekola, M. Perarnau-Llobet, U. Poschinger, A. Rolandi, D. Rosa, R. Sánchez, A. C. Santos, R. S. Sarthour, E. Sela, A. Solfanelli, A. M. Souza, J. Splettstoesser, D. Tan, L. Tesser, T. Van Vu, A. Widera, N. Yunger Halpern, and K. Zawadzki, Roadmap on quantum thermodynamics, Quantum Science and Technology **11**, 012501 (2026).

[7] F. Binder, L. A. Correa, C. Gogolin, J. Anders, and G. Adesso, eds., *Thermodynamics in the Quantum Regime: Fundamental Aspects and New Directions* (Springer, Cham, 2018).

[8] C. Jarzynski, Nonequilibrium work theorem for a system strongly coupled to a thermal environment, Journal of Statistical Mechanics: Theory and Experiment **2004**, P09005 (2004).

[9] H. J. D. Miller and J. Anders, Entropy production and time asymmetry in the presence of strong interactions, Phys. Rev. E **95**, 032123 (2017).

[10] A. M. Lacerda, A. Purkayastha, M. Kewming, G. T. Landi, and J. Goold, Quantum thermodynamics with fast driving and strong coupling via the mesoscopic leads approach, Phys. Rev. B **107**, 195117 (2023).

[11] F. Binder, S. Vinjanampathy, K. Modi, and J. Goold, Quantum thermodynamics of general quantum processes, Physical Review E **91**, 032119 (2015).

[12] P. Pechukas, Reduced Dynamics Need Not Be Completely Positive, Physical Review Letters **73**, 1060 (1994).

[13] M. F. Gelin and M. Thoss, Thermodynamics of a subensemble of a canonical ensemble, Physical Review E **79**, 051121 (2009).

[14] P. Hänggi, G.-L. Ingold, and P. Talkner, Finite quantum dissipation: The challenge of obtaining specific heat, New Journal of Physics **10**, 115008 (2008).

[15] G.-L. Ingold, P. Hänggi, and P. Talkner, Specific heat anomalies of open quantum systems, Physical Review E **79**, 061105 (2009).

[16] M. Esposito, M. A. Ochoa, and M. Galperin, Nature of heat in strongly coupled open quantum systems, Physical Review B **92**, 235440 (2015).

[17] Á. Rivas, Strong Coupling Thermodynamics of Open Quantum Systems, Physical Review Letters **124**, 160601 (2020).

[18] P. Strasberg and M. Esposito, Measurability of nonequilibrium thermodynamics in terms of the Hamiltonian of mean force, Physical Review E **101**, 050101 (2020).

[19] L. A. Correa and J. Glatthard, Potential renormalisation, Lamb shift and mean-force Gibbs state – to shift or not to shift?, Quantum **9**, 1868 (2025).

[20] M. Campisi, P. Talkner, and P. Hänggi, Thermodynamics and fluctuation theorems for a strongly coupled open quantum system: An exactly solvable case, Journal of Physics A: Mathematical and Theoretical **42**, 392002 (2009).

[21] A. O. Caldeira and A. J. Leggett, Quantum tunnelling in a dissipative system, Annals of Physics **149**, 374 (1983).

[22] H. Grabert, P. Schramm, and G.-L. Ingold, Quantum Brownian motion: The functional integral approach, Physics Reports **168**, 115 (1988).

[23] S. Hilt, B. Thomas, and E. Lutz, Hamiltonian of mean force for damped quantum systems, Physical Review E **84**, 031110 (2011).

[24] A. J. Leggett, S. Chakravarty, A. T. Dorsey, M. P. A. Fisher, A. Garg, and W. Zwerger, Dynamics of the dissipative two-state system, Reviews of Modern Physics **59**, 1 (1987).

[25] J. D. Cresser and J. Anders, Weak and Ultrastrong Coupling Limits of the Quantum Mean Force Gibbs State, Physical Review Letters **127**, 250601 (2021).

[26] K. V. Hovhannisyan, S. Nemati, C. Henkel, and J. Anders, Hamiltonian of mean force for strongly-coupled systems, Physical Review E **102**, 012104 (2020).

[27] X. Du, J. Wang, and Y. Ma, Generalized Hamiltonian of mean-force approach to open quantum systems coupled to finite baths in thermoequilibrium, Physical Review E **111**, 024139 (2025).

[28] P. C. Burke, G. Nakerst, and M. Haque, Structure of the Hamiltonian of mean force, Physical Review E **110**, 014111 (2024).

[29] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, Oxford, 2002).

[30] A. Nazir and G. Schaller, The reaction coordinate mapping in quantum thermodynamics, Thermodynamics in the Quantum Regime , 551 (2018).

[31] J. Iles-Smith, A. G. Dijkstra, N. Lambert, and A. Nazir, Energy transfer in structured and unstructured environments: Master equations beyond the born–markov approximations, The Journal of Chemical Physics **144**, 044110 (2016).

[32] N. Anto-Sztrikacs, A. Nazir, and D. Segal, Effective-hamiltonian theory of open quantum systems at strong coupling, PRX Quantum **4**, 020307 (2023).

[33] M. Shubrook, J. Iles-Smith, and A. Nazir, Non-markovian quantum heat statistics with the reaction coordinate mapping, Quantum Science and Technology **10**, 025063 (2025).

[34] J. M. Moix, Y. Zhao, and J. Cao, Equilibrium-reduced density matrix formulation: Influence of noise, disorder, and temperature on localization in excitonic systems, Physical Review B **85**, 115412 (2012).

[35] X. Chen, The rigorous stochastic matrix multiplication scheme for the calculations of reduced equilibrium density matrices of open multilevel quantum systems, The Journal of Chemical Physics **140**, 154101 (2014).

[36] N. Makri, Exploiting classical decoherence in dissipative quantum dynamics: Memory, phonon emission, and the blip sum, Chemical Physics Letters **593**, 93 (2014).

[37] Y. Tanimura, Reduced hierarchical equations of motion in real and imaginary time: Correlated initial states and thermodynamic quantities, The Journal of Chemical Physics **141**, 044114 (2014).

[38] L. Song and Q. Shi, Calculation of correlated initial state in the hierarchical equations of motion method using an imaginary time path integral approach, The Journal of Chemical Physics **143**, 194106 (2015).

[39] M. Xu, V. Vadimov, J. T. Stockburger, and J. Ankerhold, Simulating non-markovian dynamics in open quantum systems, arXiv preprint 10.48550/arXiv.2601.02160 (2026), comment: Rev. Mod. Phys (submitted), v2 7 Jan 2026, arXiv:2601.02160 [quant-ph].

[40] J. T. Stockburger, Simulating spin-boson dynamics with stochastic Liouville–von Neumann equations, Chemical Physics **296**, 159 (2004).

[41] G. M. G. McCaul, C. D. Lorenz, and L. Kantorovich, Partition-free approach to open quantum systems in harmonic environments: An exact stochastic Liouville equation, Physical Review B **95**, 125124 (2017).

[42] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, Annals of Physics **24**, 118 (1963).

[43] J. Hubbard, Calculation of Partition Functions, Physical Review Letters **3**, 77 (1959).

[44] R. L. Stratonovich, On a method of calculating quantum distribution functions, Soviet Physics Doklady **2**, 416 (1957).

[45] J. T. Stockburger and H. Grabert, Exact c -Number Representation of Non-Markovian Quantum Dissipation, Physical Review Letters **88**, 170407 (2002).

[46] H. Kleinert and S. V. Shabanov, Quantum Langevin equation from forward-backward path integral, Physics Letters A **200**, 224 (1995).

[47] N. G. van Kampen, Derivation of the quantum Langevin equation, Journal of Molecular Liquids Proceedings of the XXV Winter Meeting on Statistical Physics, **71**, 97 (1997).

[48] W. Magnus, On the exponential solution of differential equations for a linear operator, Communications on Pure and Applied Mathematics **7**, 649 (1954).

[49] S. Blanes, F. Casas, J. Oteo, and J. Ros, The Magnus expansion and some of its applications, Physics Reports **470**, 151 (2009).

[50] J. Wei and E. Norman, Lie Algebraic Solution of Linear Differential Equations, Journal of Mathematical Physics **4**, 575 (1963).

[51] B. L. Hu, J. P. Paz, and Y. Zhang, Quantum Brownian motion in a general environment: Exact master equation with nonlocal dissipation and colored noise, Physical Review D **45**, 2843 (1992).

[52] G. McCaul and Denys. I. Bondar, How to win friends and influence functionals: Deducing stochasticity from deterministic dynamics, The European Physical Journal Special Topics **230**, 733 (2021).

[53] R. Kubo, Generalized cumulant expansion method, Journal of the Physical Society of Japan **17**, 1100 (1962).

[54] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

[55] A. Das, *Finite Temperature Field Theory* (World Scientific, 1997).

[56] U. Weiss, *Quantum Dissipative Systems*, 4th ed. (World Scientific, Singapore, 2012).

[57] S. Nemati, C. Henkel, and J. Anders, Coupling function from bath density of states, Europhysics Letters **139**, 36002 (2022).

[58] A. Van-Brunt and M. Visser, Special-case closed form of the baker-campbell-hausdorff formula, arXiv preprint (2015), arXiv:1501.02506.

[59] H. J. D. Miller, Hamiltonian of mean force for strongly-coupled systems, in *Thermodynamics in the Quantum Regime: Fundamental Aspects and New Directions*, edited by F. Binder, L. A. Correa, C. Gogolin, J. Anders, and G. Adesso (Springer International Publishing, 2018) pp. 531–549.